

STOCHOS DIM-GP: Benchmark Results Across Tabular Learning, Physics Simulation and Optimization

Probabilistic AI for engineering, evaluated on workstation hardware

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Overview

Engineering teams work with material measurements, design parameters, simulation fields and physical tests. Turning that information into a better design requires several connected steps: predict performance, understand which inputs matter, compare alternatives and choose the next simulation or experiment. The cost of each evaluation limits how many designs a team can explore.

STOCHOS brings **probabilistic modeling, sensitivity analysis and Bayesian optimization** into one engineering environment. Its DIM-GP model family covers tabular regression and classification, geometry-based field prediction and transient simulation. Predictions include uncertainty estimates that can inform both design assessment and the selection of the next evaluation.

Why we take a workstation-focused approach

STOCHOS supports foundation models and agentic AI workflows within a platform designed for affordable workstation hardware. STOCHOS Flow makes modeling and optimization accessible through visual workflows, connects to existing simulation tools and Python, and supports sharing trained models through prediction applications.

Our focus is the combination: a probabilistic modeling platform that can serve materials, simulation and design teams, incorporate their physical knowledge, and operate within their own computing environment.

Why these benchmarks matter

A benchmark gives competing methods a common problem, evaluation data and scoring rules. We evaluated DIM-GP across **fourteen benchmark tasks and studies**, covering tabular learning, fluid dynamics, structural mechanics and optimization. The series combines externally scored competitions with evaluations against published baselines and a virtual formulation study. We publish the results with named comparisons, evaluation conditions and computing requirements so customers can assess STOCHOS’s capabilities across a broad range of tasks.

Selected results

The strongest results span several application classes. In the [PLAID](#) snapshots of 4 September 2026 (7 September for 2D_profile), DIM-GP ranks first on **four of the six boards examined when the standalone reconstructed-solver entry is excluded**. This excludes the [PhysicsX simulator on Tensile2d](#) [44]. With all entries included, DIM-GP ranks first on three of six boards and second on Tensile2d. The comparison field includes **NPco from NP Company** and **PXTransolver from PhysicsX**.

On [AhmedML](#) and [DrivAerML](#), DIM-GP has the lowest reported errors among the ten methods listed, including **Emmi AI’s AB-UPT** [40], across all five AhmedML channels and both DrivAerML surface channels.

On [LagrangeBench](#), DIM-GP improves on the best of three published baselines in 19 of 21 dataset–metric comparisons. The baselines include **Google DeepMind’s Graph Network-based Simulator (GNS)** [34]. On DeformingPlate, DIM-GP achieves **57% lower RMSE than Google DeepMind’s MeshGraphNets** [48] and ranks second among the three methods compared under the canonical protocol (§9).

In our reconstruction of the [TabArena-Lite](#) comparison, DIM-GP ranks **third among 80 model entries, or sixth among 85 entries including AutoML systems**, in an internal comparison using published results (§4). The field includes [TabPFN-3](#) from Prior Labs, [acquired by SAP in July 2026](#). On [TALENT-300](#), it has the lowest average rank in each task type among the four models in our additional comparison. In a virtual concrete-formulation study based on public laboratory data, STOCHOS meets the modeled specification in **149 of 150 runs**, compared with 137 for BoTorch’s qNEHVI.

The reported DIM-GP models train in less than one day on a local workstation, using at most one consumer GPU per run and a GPU-memory budget below 8 GB. CPU-only configurations are identified in the benchmark sections.

The scorecard

Bold type highlights DIM-GP results and selected summary findings. Error metrics are lower-is-better unless otherwise stated.

Benchmark	DIM-GP result	Field	Interpretation
PLAID VKI-LS59 (§13)	1st	21 entries	7.9% lower aggregate error than PXTransolver.
PLAID Hyperelasticity (§14)	1st	22 entries	Approximately 6% lower aggregate error than PXTransolver.
PLAID ElastoPlastoDynamics (§16)	1st	8 entries	Lowest errors on both displacement channels.
PLAID Tensile2d (§15)	2nd	22 entries	Second overall at an aggregate error of 0.0007.
PLAID Rotor37 (§12)	3rd	16 entries	Aggregate error approximately 0.00049 versus 0.00043 for the leader: a gap of about 0.00006.
PLAID 2D_profile (§17)	3rd	22 entries	Aggregate error 0.0152 versus the leader's 0.0134: a gap of 0.0018.
TabArena-51 (§4)	3rd (internal)	80 entries	Internal reconstruction; sixth of 85 with AutoML systems.
TALENT-300 (§5)	1st	4 models	Lowest average rank on common completed subsets.
AhmedML (§10)	1st	10 methods	Lowest error on all five channels; 6.7 h training.
DrivAerML (§11)	1st	10 methods	Lowest error on both surface channels; under 15 h training.
LagrangeBench (§7)	19 of 21 cells	3 baselines	Compared with the per-cell best of three methods.
DeepMind Water-3D (§8)	8.5% higher MSE	GNS	Compared with published GNS mean; fewer gradient updates.
DeepMind DeformingPlate (§9)	2nd	3 methods	57% lower RMSE than MeshGraphNets; M4GN reports lower RMSE.
Bayesian optimization (§18)	1st in success count	6 approaches	Highest success count in the virtual study: 149/150 runs.

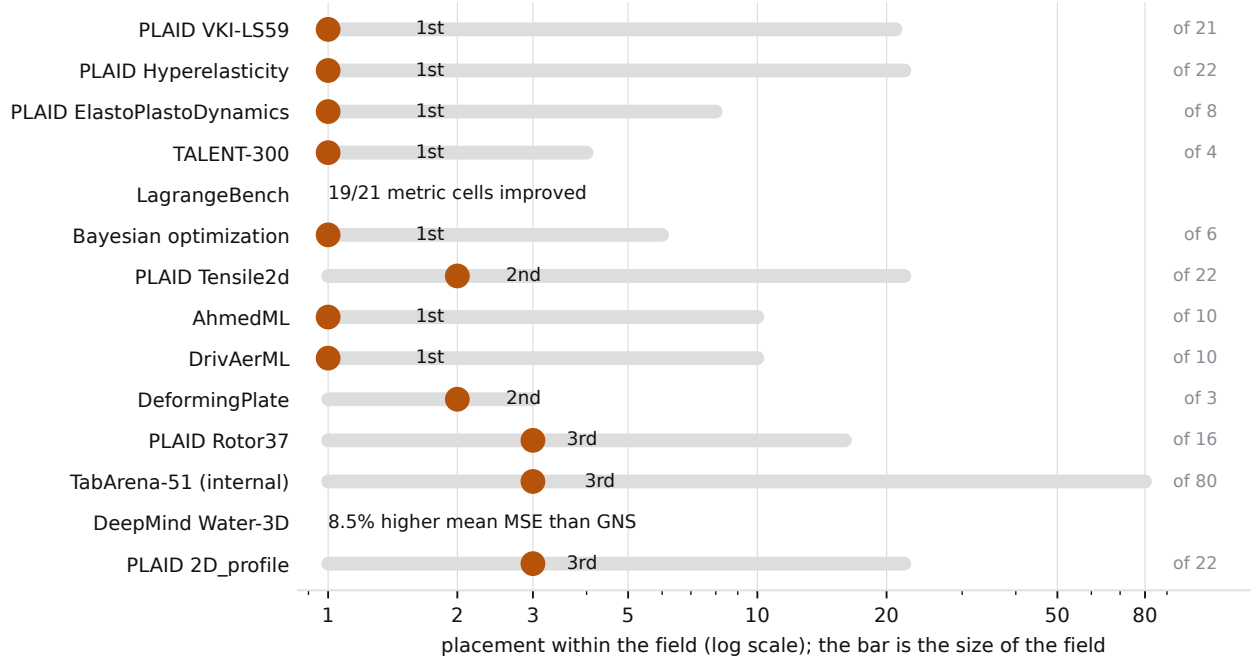


Figure 1. Comparison across the fourteen evaluated tasks and studies. Bars show the selected comparison field. Placements include public leaderboard positions and internal comparisons, as defined in the scorecard. LagrangeBench is shown by metric-cell count and Water-3D by mean error.

The scorecard counts each tabular suite once, each PLAID board separately, and the other studies individually. Across its **12 ranked comparisons**, DIM-GP outranks **203 competing benchmark entries**. Each competing entry counts once per benchmark comparison.

PLAID positions use the archived 4 September 2026 snapshots, except for 2D_profile, which uses the 7 September snapshot. Automotive results are reported as of 7 September 2026.

Computational requirements

One DrivAerML reference simulation requires approximately 61,000 CPU core-hours according to the dataset’s reported setup (§11). Once reference data are available, a trained surrogate can evaluate additional geometries at substantially lower computational cost. Figure 2 reports the training resources for the automotive comparisons.

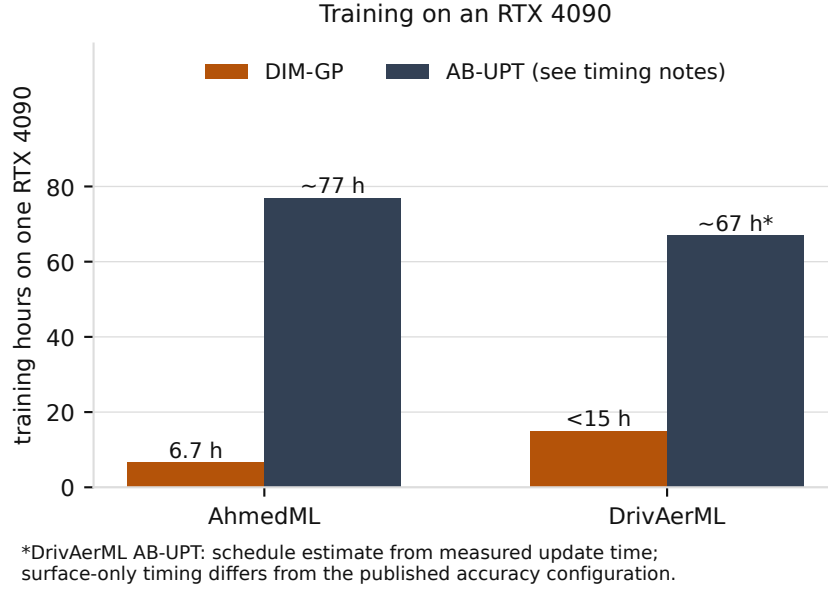


Figure 2. Training time on an RTX 4090. AhmedML times are measured runs; DrivAerML’s AB-UPT value is a schedule estimate from measured update time in a surface-only configuration (§11.1). The 15-hour DIM-GP bar is an upper bound.

Scope of the comparisons

The comparison includes named research methods and submissions from commercial vendors. On the five PLAID boards with entries from all three, DIM-GP has lower aggregate error than PXTransolver and NPco on VKI-LS59, Hyperelasticity and Tensile2d, and higher error on Rotor37 and 2D_profile. The separate PhysicsX simulator entry is reported in §15 [44].

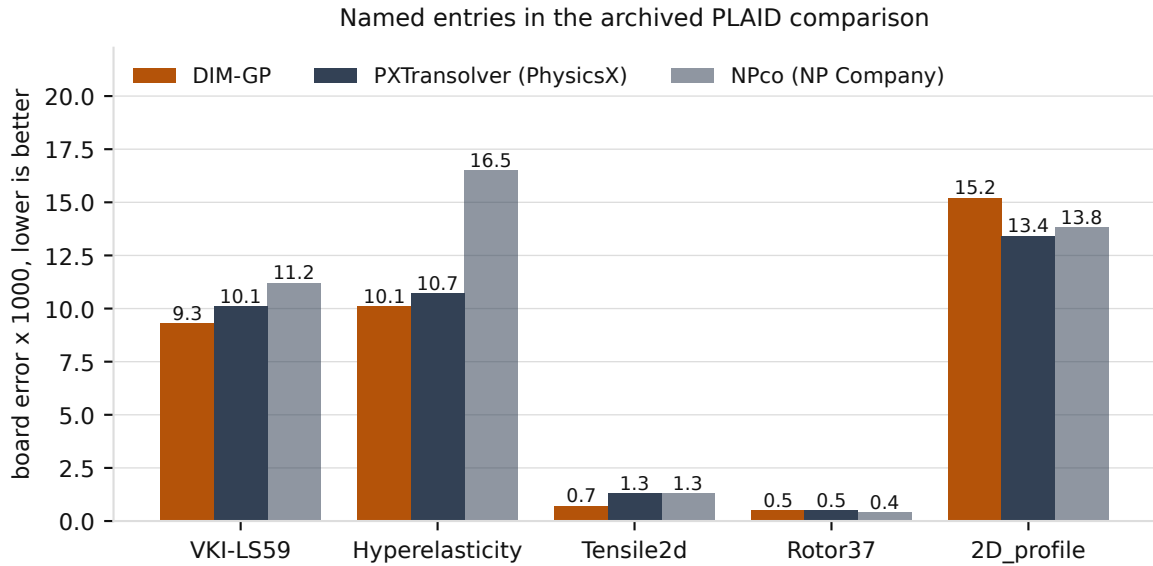


Figure 3. Aggregate leaderboard errors for the named DIM-GP, PXTransolver and NPco entries in the archived comparisons. Lower is better. The separate PhysicsX solver-based Tensile2d entry is reported in §15 and is not plotted in this three-entry comparison.

1. STOCHOS in the engineering workflow

1.1 What DIM-GP is

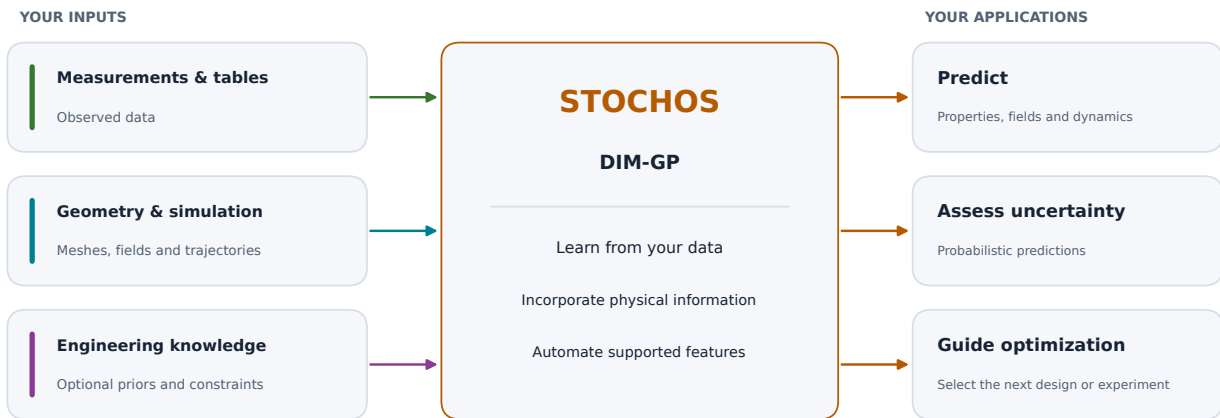
DIM-GP stands for **Deep Infinite Mixture of Gaussian Processes**, the probabilistic model family within STOCHOS. It supports predictions from tables, particle states and unstructured meshes. An engineer can use these models to estimate a material property, predict an aerodynamic field or model a transient response, with uncertainty estimates accompanying the predictions.

The tabular evaluation uses a fixed configuration across the 51 TabArena and 300 TALENT datasets, including validation-based adaptation. Physics evaluations use task-appropriate configurations. The benchmark sections report the results of these configured workflows.

1.2 Incorporating physics into DIM-GP

FROM ENGINEERING DATA TO DESIGN DECISIONS

STOCHOS capabilities at a glance



STOCHOS capabilities. Engineering data and optional physical information support prediction, uncertainty assessment and optimization. Conceptual overview of the product capabilities available across STOCHOS workflows.

DIM-GP can incorporate inexpensive physical priors, post-training conditioning on known relationships, and geometric or physical features that help it learn from engineering data. STOCHOS calculates supported geometric features automatically during mesh processing; engineers can supply application-specific physical priors and constraints. Multi-fidelity modeling also allows lower-cost simulations and higher-fidelity simulations or measurements to contribute to the same modeling workflow.

1.3 From models to engineering decisions

STOCHOS Flow connects data import, model fitting, validation, sensitivity analysis and optimization in a visual workflow. Sensitivity analysis identifies influential inputs; Bayesian optimization uses predictions and uncertainty to propose the next design or experiment. Engineers can connect Ansys Workbench or Python solvers and export prediction applications for colleagues to use. Agentic AI assists with workflow creation and troubleshooting, with local model options available.

The benchmark series evaluates the prediction and optimization methods that support this workflow.

1.4 How the results are evaluated

PLAID provides externally scored evidence. Participants receive test inputs, such as geometries and operating conditions, and submit prediction files while the reference test outputs remain withheld. The platform scores the predictions and uses hidden test subsets to discourage overfitting ([PLAID benchmark protocol](#), §4.3 and Appendix C). This allows STOCHOS to be evaluated without distributing its proprietary implementation.

The automotive and tabular sections compare our evaluations with published reference results; §4.2 explains the submission requirements behind the internal TabArena comparison. The formulation study uses a virtual test bench based on public laboratory data. Each section identifies its comparison field, metrics and evaluation conditions.

2. Tabular benchmarks and evaluation protocol

	TabArena-51	TALENT-300
Datasets	51 curated, real-world	300
Protocol	TabArena-Lite (single split, fold 0)	single seed, fixed splits
Metrics	roc_auc (30 sets), log_loss (8), rmse (13)	RMSE / accuracy
Field	79 model variants + 5 AutoML systems	28–31 published methods (official) + 3 foundation models we added
Hardware	1× RTX 4090; DIM-GP <8 GB	same

Hardware. The experiments used an Intel Core i9-13900KS workstation (24 cores, 32 threads), 64 GB of DDR5-4200 memory and NVIDIA GeForce RTX 4090 graphics under Windows 11 Pro. The workstation contains two cards, used for separate experiments; each DIM-GP run uses at most one GPU, with GPU memory below 8 GB. Published baseline timings identify their own hardware.

Our TabArena comparison uses fold 0 under the official **TabArena-Lite** protocol and the corresponding published Lite results. TALENT uses its fixed splits.

2.0 Definitions

Let $\mathcal{M}$ be the set of methods, $\mathcal{D}$ the set of datasets, and $e_{m,d}$ the error of method m on dataset d under that dataset’s own metric — one minus ROC-AUC on the 30 binary sets, log-loss on the 8 multiclass sets, RMSE on the 13 regression sets. Lower is better throughout. Because these three quantities share no scale, no statistic below averages $e_{m,d}$ across datasets directly.

Elo. TabArena fits a Bradley–Terry model [29] to pairwise wins across datasets. The model assigns method m a probability of beating method m' :

$$P(m \succ m') = \frac{1}{1 + B^{(R_{m'} - R_m)/S}}, \quad B = 10, S = 400$$

Reported Elo is the median over 200 task-level bootstrap rounds, with confidence bounds at the 2.5% and 97.5% quantiles. Elo reflects who wins; normalized error also reflects the size of error differences. Section 4.2 reports both, together with mean rank.

Normalized error. Per dataset, the error is rescaled against the best and the *median* method on that dataset, then clipped:

$$\tilde{e}_{m,d} = \text{clip} \left(\frac{e_{m,d} - \min_{m'} e_{m',d}}{\max(\text{med}_{m'} e_{m',d} - \min_{m'} e_{m',d}, 10^{-5})}, 0, 1 \right), \quad \tilde{e}_m = \frac{1}{|\mathcal{D}|} \sum_d \tilde{e}_{m,d}$$

Here, 0 denotes the best observed method and 1 the median or a worse result. Clipping limits the influence of a single unusually large error on the aggregate.

Headroom (§2.1) measures how much a dataset separates methods at all, using the same two quantities as that denominator:

$$h_d = \frac{\text{med}_m e_{m,d} - \min_m e_{m,d}}{\text{med}_m e_{m,d}}$$

$h_d \rightarrow 0$ means the median error approaches the best observed error. Larger values indicate greater separation between the best and median.

Cost. Training and inference costs normalize the complete fit and prediction pass by the number of rows, then take the median across datasets:

$$c_m^{\text{fit}} = \text{med}_d \frac{1000 t_{m,d}^{\text{fit}}}{n_d^{\text{train}}}, \quad c_m^{\text{inf}} = \text{med}_d \frac{1000 t_{m,d}^{\text{inf}}}{n_d^{\text{test}}}$$

These measures include fixed per-dataset overhead. Section 4.5 estimates fixed and per-row contributions with a robust Theil–Sen fit [30], $t_{m,d}^{\text{fit}} \approx a_m + b_m n_d^{\text{train}}$.

2.1 What is in TabArena-51

TabArena contains 51 curated real-world datasets [1], checked for leakage, duplication and mislabelled task types. Each dataset has a fixed evaluation metric (Figure 4).

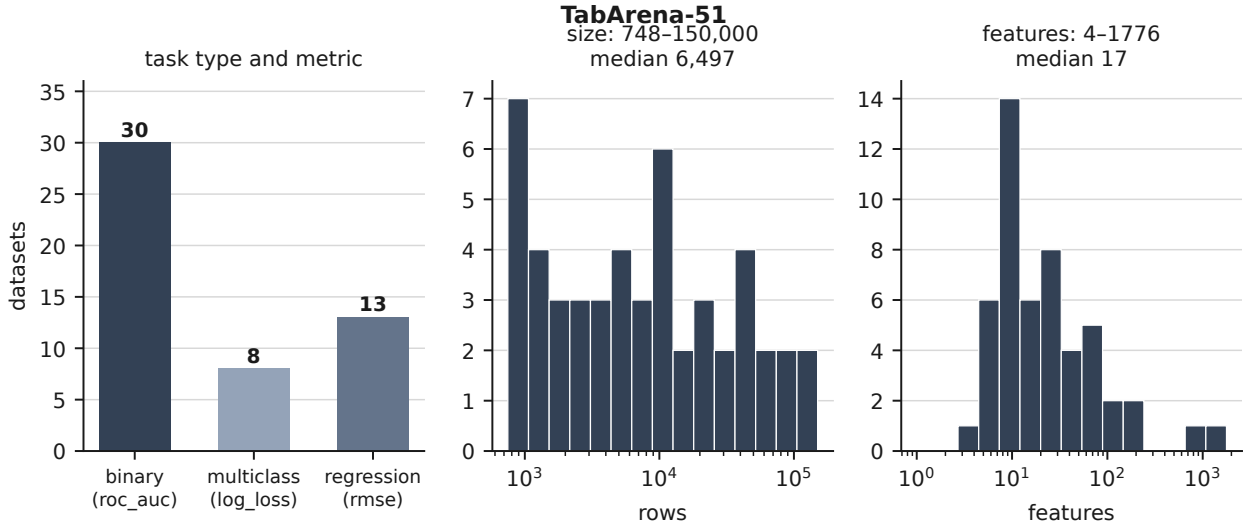


Figure 4. TabArena-51 at a glance: task type and metric, the distribution of dataset sizes, and the distribution of feature counts. The benchmark is three sub-benchmarks of unequal size — 30 binary sets scored on ROC-AUC, 8 multiclass on log-loss, 13 regression on RMSE.

Task	Datasets	Metric	Rows (median)	Rows (range)	Features (median)
Binary	30	roc_auc	9,911	748 – 150,000	21
Multiclass	8	log_loss	2,445	898 – 78,053	37
Regression	13	rmse	6,497	907 – 53,940	9

The suite spans 748 to 150,000 rows and 4 to 1,776 features. Binary classification accounts for 30 of the 51 datasets and uses ROC-AUC, a measure of predicted ordering.

Where the data comes from, and how hard it is

TabArena covers commercial and scientific applications; business, marketing and finance account for nearly half the datasets. Figure 5 and the table below summarize their domains and the median-to-best error gaps defined in §2.0.



Figure 5. TabArena dataset domains, training-set shapes and median-to-best error gaps. The diagonal in the centre panel marks equal numbers of training rows and features.

Field	Datasets	Training rows	Median headroom
Business & marketing	16	1,000 – 86,586	12.3 %
Finance	8	666 – 100,000	5.8 %
Chemistry & material science	6	598 – 30,486	10.5 %
Medical & healthcare	6	498 – 47,678	10.5 %
Biology & life sciences	5	604 – 2,500	15.0 %
Technology & internet	4	902 – 7,256	15.3 %
Physics & astronomy	3	1,002 – 52,035	24.9 %
Industry & manufacturing	1	50,666	51.9 %
Environmental science & climate	1	1,722	9.1 %
Education	1	2,949	6.9 %

The suite includes imbalanced industrial and credit tasks, as well as categorical inputs. These characteristics make the stated metric and consistent preprocessing important to the comparison.

2.2 What is in TALENT-300

TALENT provides 300 datasets [2]: 119 regression, 101 binary and 80 multiclass, spanning 252 to 634,460 rows (median 3,389) and 3 to 970 features (median 16), across 15 application domains — finance and economics (53 datasets), healthcare (31), image and signal (28), industrial and sensors (27), biology and genomics (21), and others.

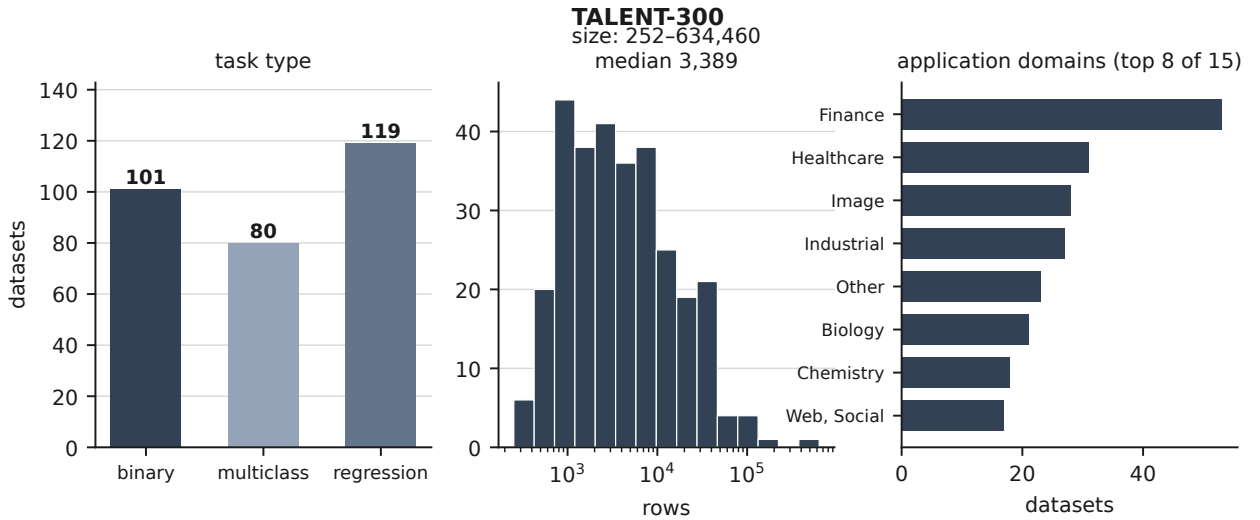


Figure 6. TALENT-300: task types, dataset sizes and application domains across 300 datasets, spanning 252 to 634,460 rows.

TabArena emphasizes curated comparisons against an actively maintained field. TALENT extends the evaluation across more datasets and application domains.

3. Tabular model families and competitive context

3.1 The three families in the field

The comparison includes three established families: gradient-boosted trees, neural networks trained per dataset and pretrained tabular models.

Gradient-boosted decision trees — LightGBM [18], CatBoost [19] and XGBoost [20] — are widely used tabular methods that fit per dataset and support CPU execution. The comparison includes both default and tuned configurations.

Neural networks for tabular data include RealMLP [21], TabM [22], ModernNCA [23] and xRFM [24]. RealMLP uses defaults developed across multiple datasets and is the highest-ranked non-pretrained entry in this comparison, at tenth place.

Tabular foundation models learn across tasks before adapting to a new dataset. The field includes TabPFN [5, 6] and its variants, TabICL [7], TabFM [3], EXAONE-Tabular [4], TabDPT [9], LimiX [8], Mitra [10] and other published models [11–17]. TabFM and EXAONE-Tabular lead the TabArena comparison presented here.

3.2 Parameter counts and interpretation

The released configurations illustrate the range of model sizes: TabFM [3] contains 1,639,444,298 parameters (1.64 billion) and EXAONE-Tabular [4] contains 20,807,434. Their TabArena-Lite Elo ratings in this comparison are 1793 and 1749, respectively.

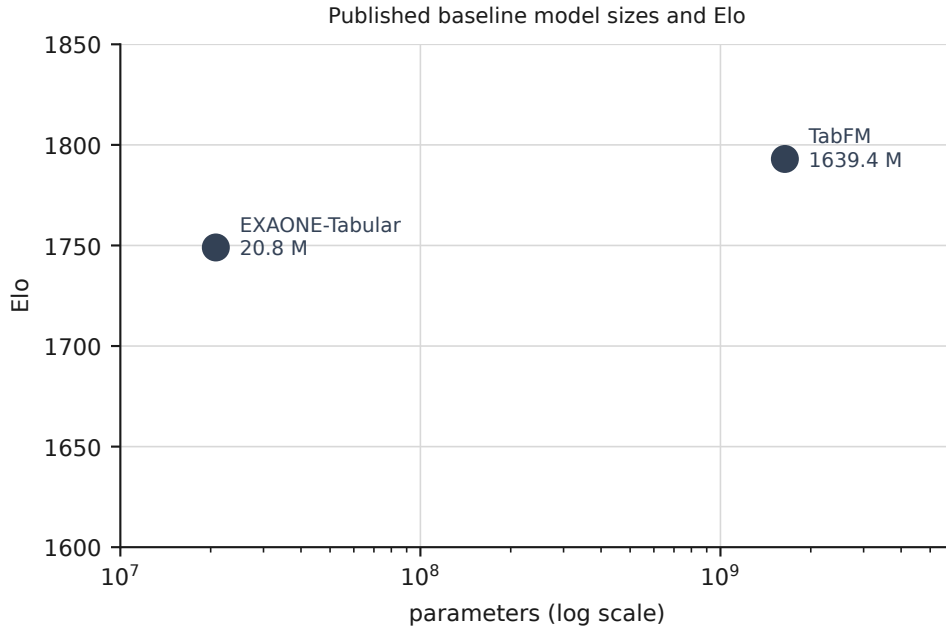


Figure 7. Parameter counts and TabArena Elo for two published baselines, from their released configurations. The models use different training data and computational budgets.

4. TabArena-51

DIM-GP ranks **third among 80 model entries** by Elo, mean rank and normalized error in our internal reconstruction of TabArena-Lite. Including AutoML systems places it sixth among 85 entries. The following sections describe the scoring check, comparison field and computational cost.

4.1 Scoring validation

Our comparison was rebuilt from TabArena’s published per-method results and compared against the official `splits_lite` CSVs:

Method	official (lite)	ours	Δ
TabFM+	1836	1840.9	+4.9
AutoGluon 1.6 (noncommercial)	1831	1832.5	+1.5
TabFM (default)	1796	1799.8	+3.8
AutoGluon 1.6 (extreme)	1757	1757.9	+0.9
EXAONE-Tabular (default)	1746	1744.7	−1.3
AutoGluon 1.5 (extreme)	1646	1645.1	−0.9
TabPFN-3 (default)	1639	1638.5	−0.5
TabPFN-2.6 (default)	1604	1603.2	−0.8
TabICLv2 (default)	1555	1555.7	+0.7

The selected reference ratings are reproduced within 4.9 Elo points. The official comparison data were obtained from the leaderboard Space’s published results.

4.2 Standing — *internal*

Comparison status. TabArena’s [submission protocol](#) requires executable training and evaluation code integrated into its benchmark. STOCHOS is licensed commercial software and was not provided through that integration route. Its results are therefore reported as an internal comparison with TabArena’s published results.

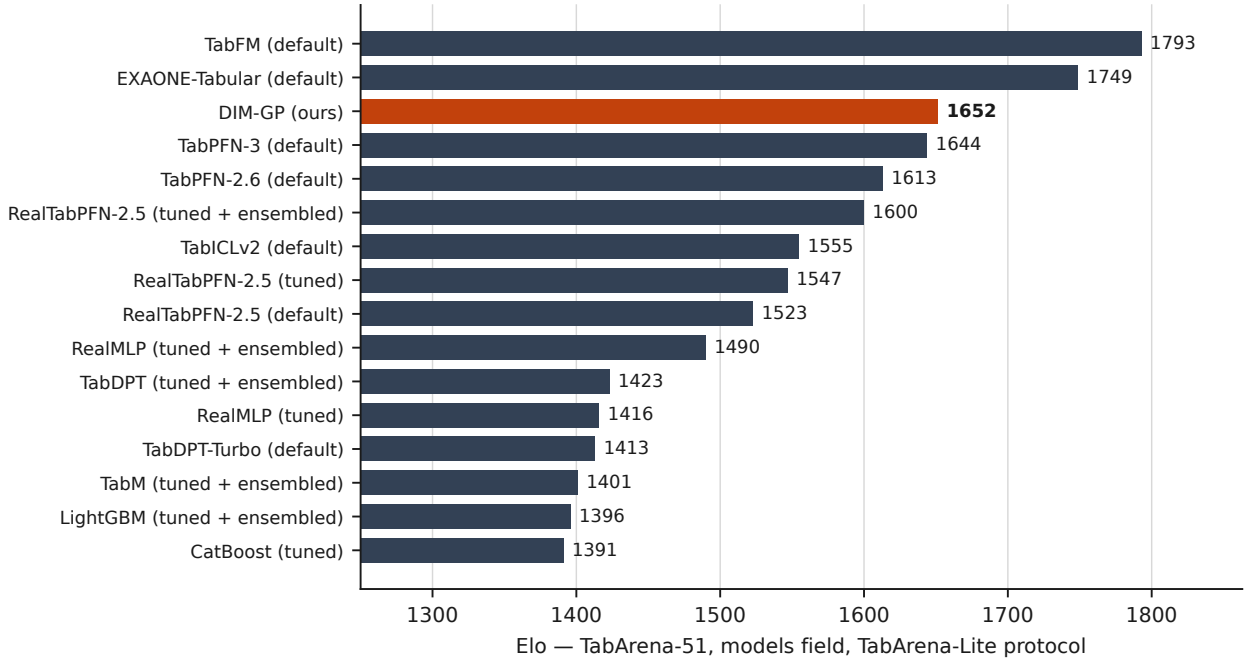


Figure 8. The TabArena-51 models leaderboard under the Lite protocol, AutoML systems excluded. Elo as defined in §2.0; DIM-GP scored into the official pool by the pipeline validated in §4.1.

Selected positions from the models field (AutoML systems excluded):

#	Model	Overall	Class.	Regr.	Binary	Multi.	Small	Medium
1	TabFM	1793	1752	2168	1780	1699	1785	1911
2	EXAONE-Tabular	1749	1747	1919	1756	1752	1698	2021
3	DIM-GP	1652	1635	1861	1654	1606	1649	1751
4	TabPFN-3	1644	1617	1897	1634	1594	1624	1789
5	TabPFN-2.6	1613	1590	1847	1585	1645	1596	1747
6	RealTabPFN-2.5 (t+e)	1600	1573	1849	1547	1745	1595	1700
7	TabICLv2	1555	1554	1696	1573	1520	1536	1691
10	RealMLP (t+e)	1490	1475	1666	1489	1451	1476	1608
16	CatBoost (tuned)	1391	1388	1494	1378	1457	1354	1561

All entries are (default) configurations unless marked (t+e) = tuned + ensembled. Rows 1–7 are foundation models, 10 a neural network, 16 tree-based.

With systems included, DIM-GP is 6th of 85 at 1647, just above AutoGluon 1.5 extreme (1646) and TabPFN-3 (1639); AutoGluon 1.6 (1757/1831) and TabFM+ (1836) rank above.

Agreement across ranking measures

Mean rank averages each method’s position across datasets; normalized error also reflects the size of error differences (§2.0). Both give DIM-GP the same placement as Elo:

#	Model	Elo	Mean rank	Normalized error	Win rate
1	TabFM	1793	6.18	0.1860	0.934
2	EXAONE-Tabular	1749	7.66	0.2501	0.916
3	DIM-GP	1652	11.27	0.3902	0.870
4	TabPFN-3	1644	11.66	0.3909	0.865
5	TabPFN-2.6	1613	13.18	0.4384	0.846
6	RealTabPFN-2.5 (t+e)	1600	13.83	0.4483	0.838
7	TabICLv2	1555	16.24	0.4728	0.807

DIM-GP is third under all three measures, with the same ordering of the top nine entries. The margin to TabPFN-3 is small: approximately eight Elo points and normalized errors of 0.3902 versus 0.3909. The paired analysis below assesses the differences across datasets.

Split by task type, the picture matches §4.3:

	Place	Normalized error	Directly ahead
Classification (38 datasets)	3 of 78	0.4234	EXAONE 0.2543
Regression, rmse (13 datasets)	5 of 77	0.3256	TabPFN-3 0.2966, Nori-30M 0.3183

Against TabPFN-3, DIM-GP has lower normalized error on classification (0.4234 versus 0.4334) and higher error on regression. TabArena contains 13 regression datasets; the complementary TALENT comparison covers 116 (§5.2).

Statistical comparisons

The paired analysis compares methods on the same 51 datasets using a Wilcoxon signed-rank test [31] on normalized-error differences and an exact sign test on wins and losses. Negative differences favour DIM-GP.

Rival	W-L	Mean Δ	Wilcoxon p	Sign p	Conclusion
LightGBM (tuned + ens.)	44-7	-0.353	<0.001	<0.001	DIM-GP better
CatBoost (tuned + ens.)	39-12	-0.338	<0.001	0.0002	DIM-GP better
LimiX	40-10	-0.283	<0.001	<0.001	DIM-GP better
TabICLv2	30-20	-0.076	0.059	0.203	no significant difference detected
TabPFN- 2.6	32-19	-0.045	0.124	0.092	no significant difference detected
TabPFN- 3	27-23	+0.001	0.730	0.672	not distinguishable
EXAONE- Tabular	22-29	+0.140	0.011	0.401	mixed
TabFM	9-41	+0.204	<0.001	<0.001	TabFM better

A Friedman test over thirteen leading methods rejects equal performance ($\chi^2 = 254.9$, $p = 1.3 \times 10^{-47}$). The Nemenyi post-hoc comparison [32] uses a critical distance of 2.56 average-rank points at $\alpha = 0.05$ (Figure 9).

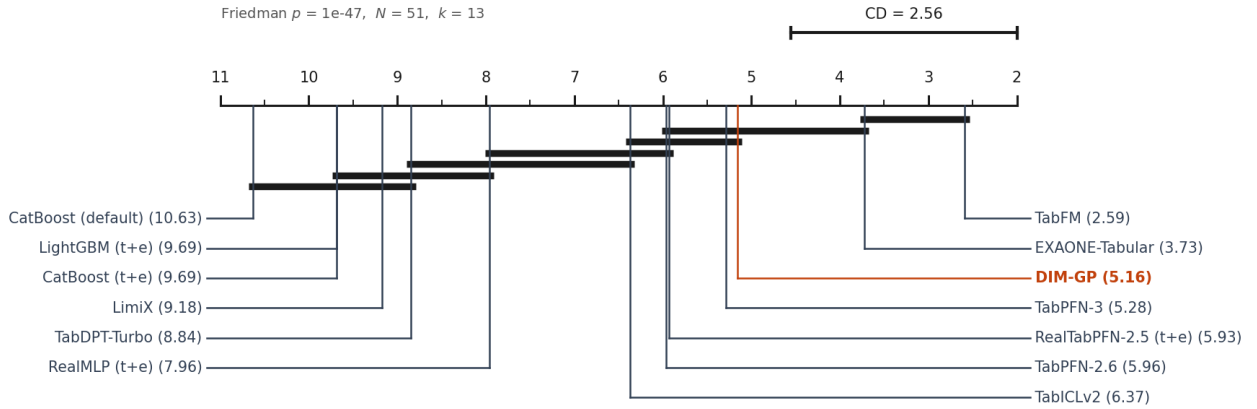


Figure 9. Critical-difference diagram (Friedman test, Nemenyi post-hoc, $\alpha = 0.05$) over the thirteen leading methods on all 51 datasets. Better average ranks lie to the right; bars join methods that the critical distance of 2.56 rank points cannot separate. Only TabFM is significantly better than DIM-GP. Ranks are computed within these thirteen methods and therefore differ from ranks over the full field of 80.

Within this thirteen-method comparison, DIM-GP’s average rank is 5.16 and **only TabFM is significantly ahead of DIM-GP under the Nemenyi analysis**. RealMLP, TabDPT-Turbo, LimiX and both tuned tree ensembles are significantly behind. Neither paired test detects a significant difference between DIM-GP and TabPFN-3.

Pairwise p-values are unadjusted across the eight comparisons; the Nemenyi procedure controls its thirteen-method comparison. EXAONE-Tabular differs under the unadjusted Wilcoxon test but falls within the Nemenyi critical distance. The paired table and model-only aggregate table use different normalization pools, so their mean-error differences are computed separately.

4.3 Per-axis reading

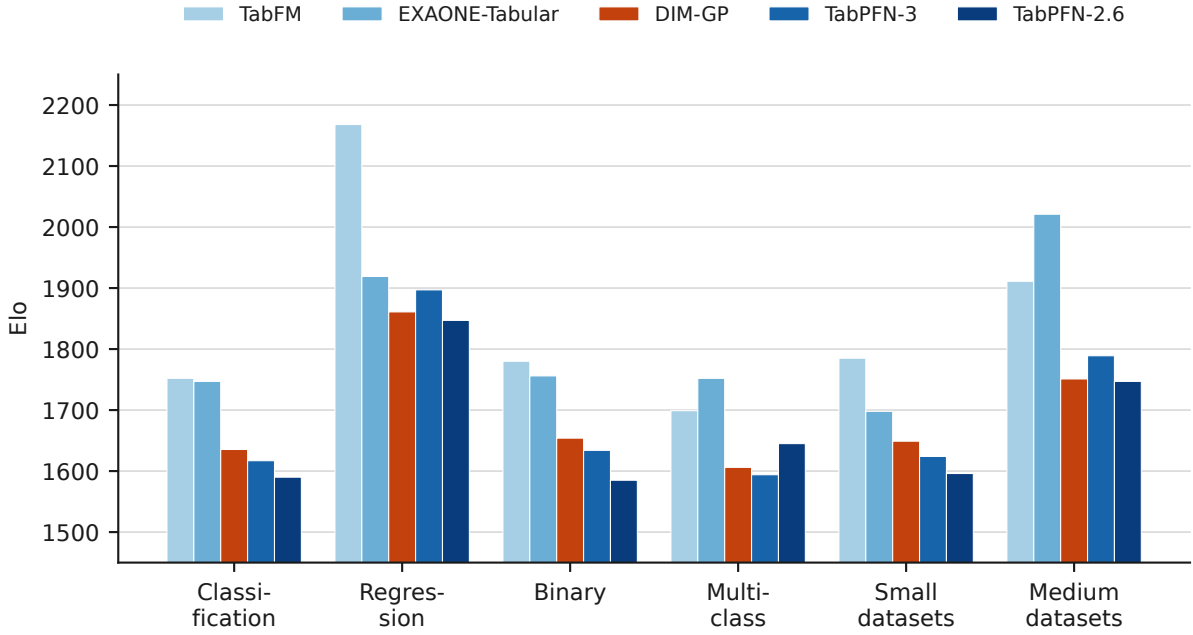


Figure 10. Elo by task type and by dataset size for the leading methods. Each axis describes a different subset of the evaluation.

Selected task-wise comparisons against TabPFN-3:

Axis	DIM-GP	TabPFN-3	Δ
Classification	1635	1617	+18
Binary	1654	1634	+20
Small datasets	1649	1624	+25
Multiclass	1606	1594	+12
Regression	1861	1897	-36
Medium datasets	1751	1789	-38

DIM-GP has higher Elo on four of the six reported axes and lower Elo on regression and medium-sized datasets in these descriptive subgroup comparisons.

4.4 Which field DIM-GP is compared in

TabArena distinguishes individual models from AutoML systems that manage model selection and training budgets. The evaluated DIM-GP configuration is fixed across datasets. Section 4.2 reports its placement in both comparison fields.

4.5 Computational cost and hardware

Using TabArena’s cost definition (§2.0), DIM-GP records median times of **30.26 seconds per 1,000 training rows** and **2.509 seconds per 1,000 prediction rows** on the RTX 4090 (51 datasets, fold 0), placing 32nd of 80 by reported fit cost. The robust timing fit separates fixed per-dataset and per-row contributions:

	Fixed cost / dataset	Per 1,000 rows	Median s/1K	measured on
TabICLv2	5.7 s	0.42 s	1.96	official board
TabPFN-3	14.2 s	0.71 s	3.59	official board
EXAONE-Tabular	8.3 s	2.43 s	6.17	official board

	Fixed cost / dataset	Per 1,000 rows	Median s/1K	measured on
DIM-GP	10.5 s	17.74 s	30.26	our RTX 4090
TabFM	33.6 s	26.00 s	40.79	official board

These fits describe complete training or conditioning procedures. Baseline timings come from the published leaderboard; DIM-GP timings are measured on our RTX 4090.

Hardware sensitivity. The official benchmark records H200 timings for TabPFN-3, TabICLv2, TabPFN-2.6 and LimiX; RTX PRO 6000 timings for TabFM and TabDPT-Turbo; and A100 timings for TabPFN-Wide. To examine the effect of hardware, we ran TabICLv2 through the official harness on our RTX 4090, using the same data splits, preprocessing and eight-fold bagging configuration:

Dataset	Rows	fit, RTX 4090	fit, H200	Factor
blood-transfusion-service-center	499	4.33 s	4.39 s	0.99
diabetes	512	4.33 s	4.09 s	1.06
qsar-biodeg	705	5.66 s	5.47 s	1.03
churn	3,400	6.24 s	5.65 s	1.10
bank-marketing	45,211	27.39 s	13.26 s	2.07
Bioresponse	3,751	173.92 s	70.21 s	2.48
APSFailure	76,000	298.29 s	107.18 s	2.78

The RTX 4090-to-H200 fit-time ratio ranges from 0.99 to 2.78 across these datasets, showing that hardware effects depend on the workload.

Applying the fitted size-dependent factor to DIM-GP gives an illustrative H200-equivalent estimate of 21.02 s/1K, versus 30.26 s/1K measured on our RTX 4090. This is not a measured DIM-GP H200 run; transferring a calibration from another model adds uncertainty.

The official entries use eight-fold bagging, while DIM-GP uses a single fit. The cost comparison reports these configurations on the shared tasks and split.

5. TALENT-300

We compare DIM-GP with TALENT’s published methods and with three additional tabular foundation models evaluated in our own runs.

5.1 Comparison with TALENT’s published methods

TALENT publishes results for 28 regression methods and 31 classification methods, including XGBoost, CatBoost, LightGBM, TabPFN, TabR, ModernNCA and RealMLP.

Task	DIM-GP avg rank	field	runner-up
Regression	2.30 — 1st of 29	90 datasets	CatBoost 6.74
Binary	4.14 — 1st of 32	95 datasets	TabR 9.14
Multiclass	3.00 — 1st of 32	77 datasets	RealMLP 6.89

DIM-GP has the lowest average rank in each task type within this published comparison field.

5.2 Additional foundation-model comparison

Our additional comparison evaluates TabPFN-3, TabICLv2 and LimiX on the TALENT datasets. Average ranks use the common set completed by all four methods, with lower values indicating better performance:

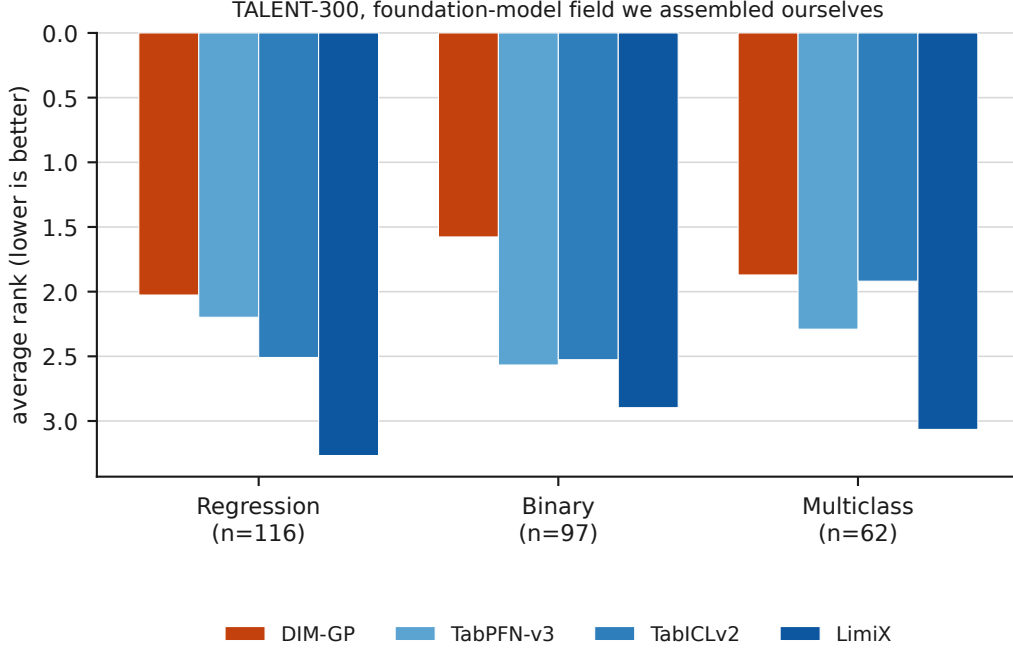


Figure 11. Average rank in our additional TALENT comparison of DIM-GP, TabPFN-3, TabICLv2 and LimiX. Each task type uses the common set of datasets completed by every method; lower is better.

Task	datasets	DIM-GP	TabPFN-3	TabICLv2	LimiX	of those, DIM-GP first
Regression	116	2.026	2.198	2.509	3.267	39
(RMSE)						
Binary	97	1.577	2.567	2.526	2.897	60
(Accu- racy)						
Multiclass	62	1.871	2.290	1.919	3.065	26
(Accu- racy)						

DIM-GP has the lowest average rank in all three task types. Section 5.3 reports completion counts for these models and for Mitra, which was excluded from the rank comparison because of its lower completion rate.

5.3 Running on one consumer GPU

The comparison uses one RTX 4090 with 24 GB of physical memory. DIM-GP’s allocation is limited to less than 8 GB. Completion counts for the tested implementations and settings are:

Model	Completed	Failed	Failure mode
DIM-GP	300 / 300	0	—
TabICLv2	298	2	OOM on the largest
TabPFN-3	295	5	OOM on the largest
LimiX	276	24	OOM on the largest
Mitra	205	95	$O(N_{\text{train}} \times N_{\text{test}})$ memory; dropped from §5.2

DIM-GP completed all 300 datasets with GPU memory below 8 GB. The baseline evaluation allows caps on the in-context sample count and ensemble size to manage resource use. The accuracy comparison in §5.2 uses the common completed datasets under those settings.

5.4 How to read these two tables

The DIM-GP and additional foundation-model evaluations use a **single seed and fixed splits**; TALENT’s published tables average multiple seeds. Classification is scored by accuracy, including threshold selection for binary tasks; TabPFN-3’s reported TALENT result uses AUC. Average ranks and win counts permit comparison across datasets with different target scales.

6. Beyond tabular data: physics surrogates

The physics evaluation covers particle dynamics, contact and large deformation, aerodynamic fields and transient structural response.

6.1 What is being evaluated

Two members of the STOCHOS model family are evaluated:

- **DIM-GP Particle** — predicts the evolution of particle and mesh-node states (§§7–9).
- **DIM-GP** — the surrogate for fields on unstructured meshes, evaluated in §§10–17.

6.2 Why these particular benchmarks

The benchmarks cover the following physical regimes and comparison fields:

Benchmark	Regime	Comparison field
LagrangeBench (§7)	Lagrangian SPH dynamics, 7 datasets	GNS, SEGNN, CoRGI
DeepMind Water-3D (§8)	Lagrangian, long-horizon rollout	GNS
DeepMind DeformingPlate (§9)	Quasi-static solid contact, tetrahedral mesh, 400 steps	MeshGraphNets and M4GN — canonical protocol
AhmedML (§10)	Steady 3D automotive aerodynamics	AB-UPT and eight other published methods
DrivAerML (§11)	Steady 3D automotive aerodynamics, high fidelity	AB-UPT and eight other published methods
PLAID Rotor37 (§12)	Parametric 3D compressor CFD	Public leaderboard
PLAID VKI-LS59 (§13)	2D transonic turbine cascade, RANS	Public leaderboard
PLAID 2D Multiscale Hyperelasticity (§14)	Finite-strain RVEs, variable topology	Public leaderboard
PLAID Tensile2d (§15)	2D elastoplastic specimen, parametric	Public leaderboard
PLAID 2D ElastoPlastoDynamics (§16)	Transient plate rupture, 41 steps	Public leaderboard
PLAID 2D_profile (§17)	Transonic airfoils, shape-only input	Public leaderboard

PLAID sections report the archived board state of 4 September 2026, except for 2D_profile, which uses the 7 September snapshot. Submissions from commercial vendors are identified by their entry names, so each comparison refers to a specific submission in that snapshot.

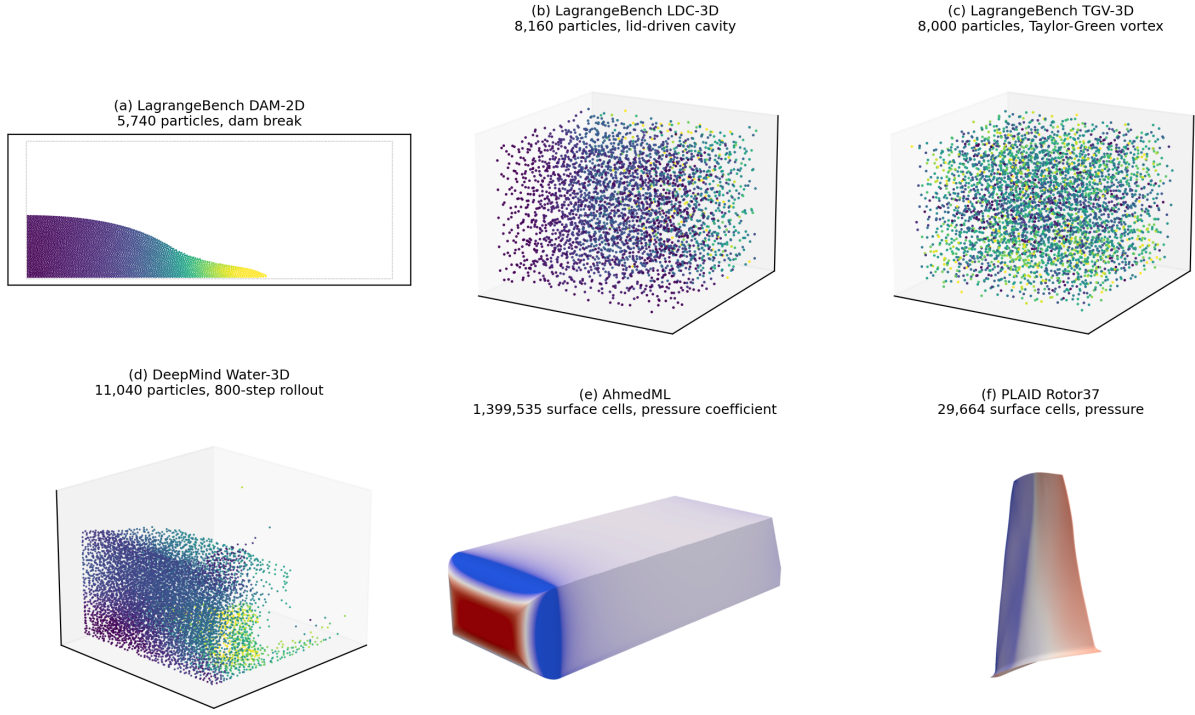


Figure 12. Reference data from four physics benchmarks. Top row: (a) dam break, (b) lid-driven cavity and (c) Taylor–Green vortex from LagrangeBench [33], followed by (d) DeepMind Water-3D [34]. Particle colours show per-step displacement; 3D cutaways expose the interior. Bottom row: (e) AhmedML [39], coloured by static pressure coefficient, and (f) PLAID Rotor37 [41], coloured by surface pressure.

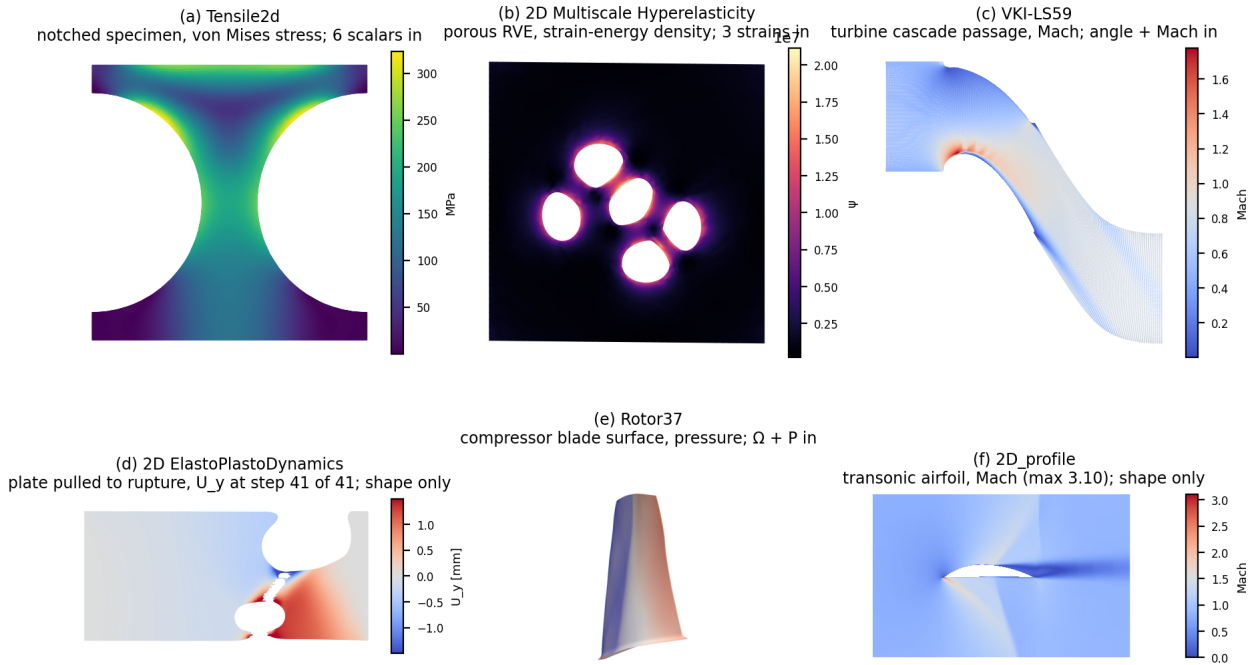


Figure 13. Reference fields from the six PLAID benchmarks (§§12–17): (a) Tensile2d, von Mises stress; (b) 2D Multiscale Hyperelasticity, strain-energy density on a deformed volume element with exaggerated displacement; (c) VKI-LS59, turbine-cascade Mach number; (d) 2D ElastoPlastoDynamics, transverse displacement at the final rupture step; (e) Rotor37, blade surface pressure; and (f) 2D_profile, airfoil Mach number.

7. LagrangeBench — *verified against the benchmark’s own evaluator*

LagrangeBench [33] is a Lagrangian-simulation benchmark: seven SPH datasets (2D and 3D Taylor–Green vortex, reverse Poiseuille flow, lid-driven cavity, and 2D dam break), a fixed protocol, and published baselines. It supports comparison across three published methods: GNS [34], SEGNN [35] and CoRGI [36] all publish numbers on the same cells under the same rules.

7.1 Protocol

Scoring follows the benchmark exactly: 26-frame windows made of 6 seed frames and 20 predicted steps, with RPF-2D scored over 384 and RPF-3D over 192 test windows. Three metrics per dataset — **MSE20** (position error over the 20 predicted steps, per dimension), **Sinkhorn** (a distributional divergence, blind to particle identity) and **Ekin** (squared error of total kinetic energy).

7.2 Result — lower error in 19 of 21 comparisons

Each DIM-GP result is compared with the **lowest published error for that dataset and metric** across GNS, SEGNN and CoRGI.

dataset	metric	best published (holder)	DIM-GP	comparison
DAM-2D	MSE20	1.55e-5 (CoRGI)	3.49e-7	44× lower error
DAM-2D	Sinkhorn	2.82e-6 (CoRGI)	6.48e-8	44× lower error
DAM-2D	Ekin	2.18e-5 (CoRGI)	8.88e-8	245× lower error
LDC-2D	MSE20	1.4e-5 (GNS)	9.34e-6	1.50× lower error
LDC-2D	Sinkhorn	5.07e-7 (CoRGI)	1.15e-7	4.39× lower error
LDC-2D	Ekin	3.81e-7 (CoRGI)	2.76e-7	1.38× lower error
RPF-2D	MSE20	1.54e-6 (CoRGI)	1.226e-6	1.26× lower error
RPF-2D	Sinkhorn	2.08e-8 (CoRGI)	1.49e-9	14× lower error
RPF-2D	Ekin	2.39e-6 (CoRGI)	7.224e-6	3.02× higher error
TGV-2D	MSE20	3.81e-6 (CoRGI)	2.29e-6	1.67× lower error
TGV-2D	Sinkhorn	1.05e-7 (CoRGI)	3.56e-8	2.95× lower error
TGV-2D	Ekin	2.90e-7 (CoRGI)	1.35e-7	2.15× lower error
LDC-3D	MSE20	3.86e-5 (CoRGI)	2.44e-5	1.58× lower error
LDC-3D	Sinkhorn	2.64e-7 (CoRGI/SEGNN)	1.48e-7	1.79× lower error
LDC-3D	Ekin	1.55e-8 (CoRGI)	1.48e-8	1.05× lower error
RPF-3D	MSE20	1.64e-5 (SEGNN)	1.330e-5	1.23× lower error
RPF-3D	Sinkhorn	1.33e-7 (CoRGI)	6.41e-8	2.08× lower error
RPF-3D	Ekin	1.34e-6 (SEGNN)	1.263e-6	parity
TGV-3D	MSE20	5.2e-3 (SEGNN)	4.76e-3	1.10× lower error

dataset	metric	best published (holder)	DIM-GP	comparison
TGV-3D	Sinkhorn	6.4e-6 (SEGNN)	3.07e-6	2.09× lower error
TGV-3D	Ekin	2.21e-3 (CoRGI/SEGNN)	1.86e-3	1.19× lower error

Compared with each method individually, DIM-GP has lower error than GNS on all 21 cells, lower error than SEGNN on 20 with one parity, and lower error than CoRGI on 19.

DIM-GP has the lowest error on all three metrics for DAM-2D, LDC-2D, TGV-2D, LDC-3D and TGV-3D.

Under the Neural-SPH 400-step protocol [37] on DAM-2D, evaluated over all 25 trajectories, the corresponding values are 3.12e-2 / 4.90e-4 / 1.99e-4 against a published best of 8.4e-2 / 7.5e-3 / 2.1e-3, a factor of 2.7 / 15 / 10.6. Since the 20-step protocol is comparatively short, this establishes that the margin persists at twenty times the horizon.

LagrangeBench: every scored cell against the best published number

DAM-2D	lower	lower	lower
LDC-2D	lower	lower	lower
LDC-3D	lower	lower	lower
RPF-2D	lower	lower	higher
RPF-3D	lower	lower	parity
TGV-2D	lower	lower	lower
TGV-3D	lower	lower	lower
	MSE20	Sinkhorn	Ekin

Figure 14. Every scored cell of LagrangeBench against the best published number for that cell, taken over GNS, SEGNN and CoRGI: seven datasets by three metrics.

7.3 The remaining two comparisons

RPF-3D Ekin — **parity**. DIM-GP records 1.263e-6 against SEGNN’s 1.34e-6 and is reported as parity.

RPF-2D Ekin — **higher error than CoRGI**. DIM-GP has 3.9-fold lower error than GNS and 2.5-fold lower error than SEGNN on this cell, while CoRGI reports the lowest error.

7.4 Cost

Each reported model was trained on a **single RTX 4090**, at roughly 20–90 seconds per epoch depending on dataset and resolution.

8. DeepMind Water-3D — *measured against the published GNS result*

Water-3D is the 3D fluid dataset from *Learning to Simulate Complex Physics with Graph Networks* [34]: approximately 14,000 particles over 800 timesteps. It tests prediction over a much longer rollout than the 20-step LagrangeBench evaluation.

8.1 Result

Scored under the paper’s own protocol — 100 test sequences, full 800-step rollouts, seed frames excluded:

model	rollout MSE (mean)	median	thickness	speed	MMD	gradient updates
GNS, as published [34]	0.01010	—	—	—	—	~20M
DIM-GP , accuracy setting	0.01096	0.00984	82%	87%	0.00265	~1.9M (10% of GNS)
DIM-GP , physics setting	0.01238	0.01109	90%	106%	0.00249	~1.6M

The accuracy setting has a mean rollout MSE 8.5% above the published GNS value, with approximately one tenth as many gradient updates. The update count describes the training schedule; wall-clock time also depends on the cost of each update.

The second setting improves the physical diagnostics, reaching 90% of reference fluid-column thickness and 106% of characteristic speed, compared with 82% and 87% for the accuracy setting. The two configurations illustrate the choice between minimizing position error and preserving bulk fluid behaviour.

8.2 Comparison scope

The comparison uses the published GNS result on Water-3D under the full 800-step protocol. LagrangeBench reports a different quantity, MSE over 20 predicted steps, while NeuralMPM [38] reports full-rollout results on 2D fluid datasets. Those results belong to their respective datasets and protocols and are evaluated separately from Water-3D.

9. DeepMind DeformingPlate — *comparison under the canonical protocol*

DeformingPlate is the solid-mechanics case of the MeshGraphNets suite [48]: a rigid actuator with a prescribed motion is pressed into a hyperelastic plate that is clamped along one edge, and the plate’s node positions are to be predicted over 400 quasi-static steps on a tetrahedral mesh of 700–2,200 nodes. The task includes contact and large deformation, and every trajectory has its own plate geometry, actuator shape and path. 1,200 trajectories are used for training, 100 for validation and 100 for the test score.

9.1 Protocol

The score follows the original paper: RMSE of world positions in metres over all nodes and all 399 predicted steps of the 100 test trajectories, with errors pooled before taking the square root and reported $\times 10^{-3}$. The seed frame is excluded. The checkpoint was selected on the validation split, and the test split was scored once.

9.2 Result

method	RMSE-all $\times 10^{-3}$	data and protocol
<i>canonical: DeepMind data, official split, original error formula</i>		
MeshGraphNets [48]	15.1	original paper
DIM-GP	6.45	this work, checkpoint chosen on the validation split
M4GN [49]	2.65	reported; no code released, not reproduced
<i>not on the canonical protocol, listed for completeness</i>		
MGN-T [50]	3.21	data regenerated with COMSOL
HCMT [51]	7.3	different error formula; its own MeshGraphNets baseline reads 7.8, DIM-GP reads 4.9 under it
ROBIN [52]	4.98	square root before the mean over trajectories; its own MeshGraphNets baseline reads 8.8
EvoMesh [53]	12.9	trained on 500 of the 1,200 trajectories
BSMS-GNN [54]	16.0	dataset re-split 1000/200/200

DIM-GP reports RMSE of 6.45 versus 15.1 for MeshGraphNets, a **57% reduction**, and ranks second among the three canonical-protocol entries. M4GN reports the lowest value, 2.65. The remaining rows use different data, splits or error conventions and are listed separately.

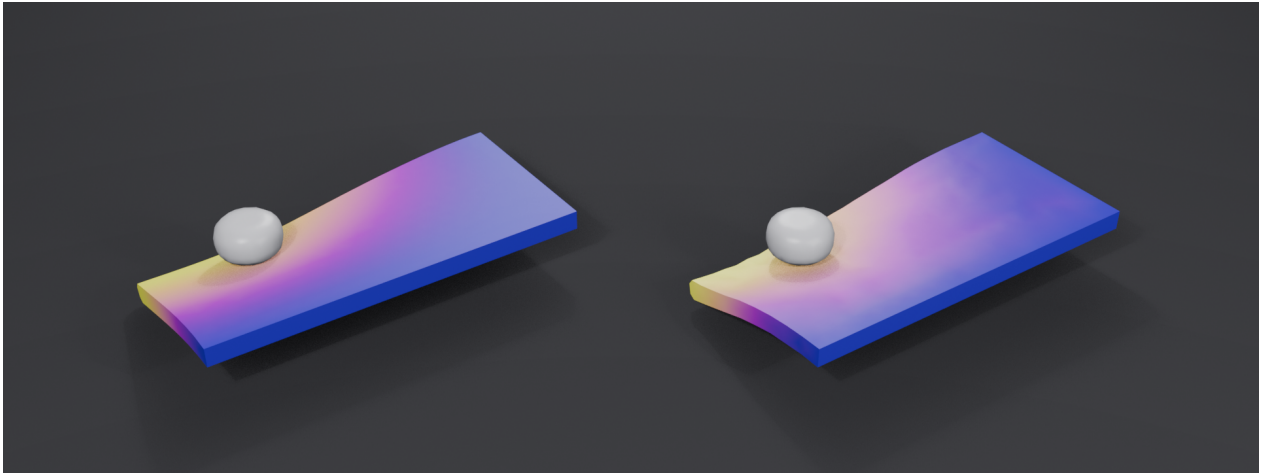


Figure 15. DeepMind DeformingPlate, test trajectory 92 at the moment of deepest indentation: the reference finite-element solution (left) and the DIM-GP rollout from the first frame alone (right), plate coloured by displacement from its rest shape, actuator in grey; the scene is turned so that the actuator approaches from above. Rollout RMSE of this trajectory 0.93×10^{-3} ; dataset mean 6.45×10^{-3} .

9.3 Cost

Training completed in **less than one day on a single RTX 4090**, with peak GPU memory **below 8 GB**.

10. AhmedML — *internal evaluation against published baselines*

AhmedML [39] is a 500-geometry CFD dataset over the Ahmed body, a standard automotive bluff-body. The published comparison in AB-UPT [40] provides the external baseline values used below.

10.1 Protocol and metric

The metric follows that comparison: per design, the Frobenius relative error $\|\hat{Y} - Y\|_2 / \|Y\|_2$, averaged over designs, reported as a percentage. Five channels are scored — surface pressure p_s and wall shear stress τ ; volume velocity u , vorticity ω and total pressure p_v . Evaluation uses raw full-resolution clouds from the 50 test designs in AB-UPT’s published split.

10.2 Result

relative L2 (%)	$p_s \downarrow$	$\tau \downarrow$	$u \downarrow$	$\omega \downarrow$	$p_v \downarrow$	training on one RTX 4090
DIM-GP [63]	2.97	3.81	1.85	6.30	1.85	6.7 h, 8 GB
AB-UPT [40]	3.01	3.88	1.90	6.52	1.98	~77 h ¹
Transformer [55]	3.41	4.03	2.09	6.76	2.16	—
Transolver [56]	3.45	4.00	2.05	8.22	2.16	—
OFormer [57]	4.12	4.60	3.63	15.06	4.08	—
UPT [58]	4.25	5.80	2.73	15.03	3.10	—
Graph U-Net [59]	6.46	7.29	4.15	53.66	5.18	—
GINO [60]	7.90	8.18	6.23	71.81	8.10	—
PointNet [61]	8.02	10.09	5.44	66.04	6.13	—
LNO [62]	12.95	11.50	7.59	72.49	8.48	—

¹ Measured for AB-UPT’s 200K-update schedule on the same RTX 4090 (§10.3). Both models use the same training data, including surface and volume fields. Other methods have no timing on this hardware.

DIM-GP places first of ten on all five channels in the comparison with published baselines [40]. Relative to AB-UPT, error is lower by 1.3% on surface pressure, 1.8% on wall shear, 2.6% on volume velocity, 3.4% on vorticity and 6.6% on total pressure.

The DIM-GP row reports our internal evaluation as of 7 September 2026. Baseline values come from the published comparison [40]. The ranking is based on point estimates.

10.3 Cost — *measured on identical hardware*

DIM-GP trains in **6.7 h on one RTX 4090 under an 8 GB cap**. We executed AB-UPT’s released code on the same hardware and measured **~77 h** for its 200K-update training schedule. Both training times are validated on our hardware, with the same data and split, surface and volume together. DIM-GP therefore requires approximately **91% less training time in this configuration**.

11. DrivAerML — *internal evaluation against published baselines*

DrivAerML [43] comprises 484 usable variants of the DrivAer notchback vehicle, each solved by hybrid RANS–LES (SA- σ -DDES, OpenFOAM) on a ~ 160 M-cell mesh at a cost of roughly 40 hours on 1,536 CPU cores per design — on the order of 61,000 core-hours per geometry. The learning task is steady: one geometry in, the time-averaged surface pressure and wall-shear-stress fields out (~ 8.8 M surface cells). Evaluation follows AB-UPT [40]: its published 400/34/50 split and per-design relative L2 error, using the Frobenius norm over the three shear components and averaging over the 50 test designs.

relative L2 (%)	surface pressure	wall shear	trained on	training on one RTX 4090
DIM-GP [63]	3.74	7.15	surface only	under 15 h
AB-UPT [40]	3.82	7.29	surface + volume	~ 67 h ²
Transformer [55]	4.35	8.26	surface + volume	—
Transolver [56]	4.81	8.95	surface + volume	—
OFormer [57]	4.85	8.92	surface + volume	—
UPT [58]	7.44	12.93	surface + volume	—
GINO [60]	13.03	21.71	surface + volume	—
Graph U-Net [59]	16.13	27.84	surface + volume	—
LNO [62]	20.51	36.44	surface + volume	—
PointNet [61]	23.63	41.85	surface + volume	—

² Schedule estimate from measured throughput of the released implementation in a surface-only configuration on our RTX 4090 (§11.1). The published AB-UPT accuracy uses surface and volume data.

DIM-GP places 1st of 10 on both channels, with 2.1% less surface-pressure error and 1.9% less wall-shear error than runner-up AB-UPT. The DIM-GP values report our internal evaluation as of 7 September 2026. Baseline values come from the published comparison [40]. The ranking is based on point estimates.

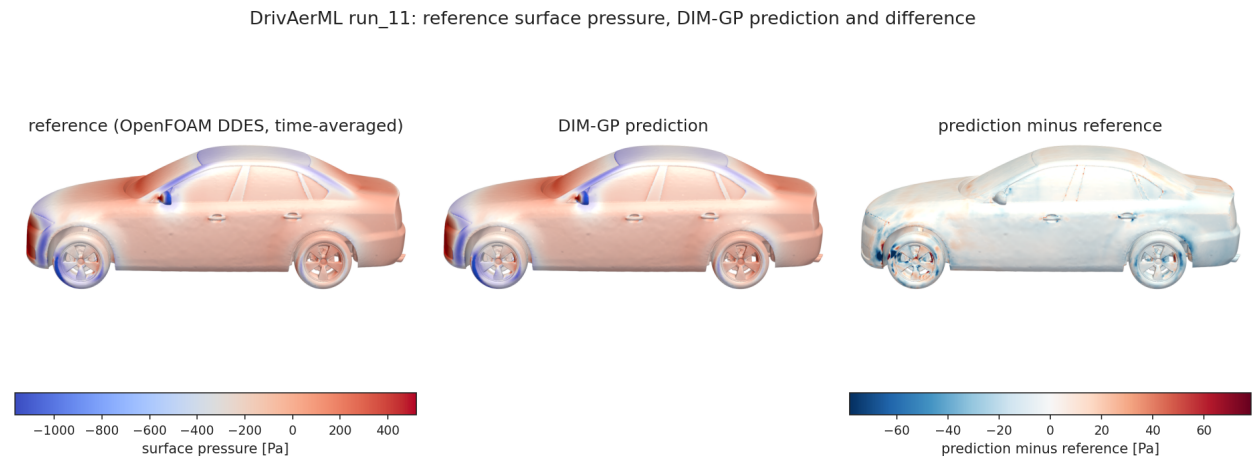


Figure 16. DrivAerML test car run_11, side view: time-averaged surface pressure of the reference simulation, the DIM-GP prediction, and their difference. Every cell of the 8.8-million-cell surface is predicted; the view shows a 400,000-cell random subset.

11.1 Computational cost on identical hardware

The released AB-UPT implementation (Noether, Emmi AI, v2026.4.0) was executed on the same RTX 4090 workstation and 400-design split in a surface-only configuration. Its measured update time of 1.21 seconds implies approximately **67 hours for 200,000 updates**. This is a schedule estimate derived from measured throughput, compared with **under 15 hours** of DIM-GP wall-clock training, including evaluation checkpointing.

The timing was measured in our Windows environment with the available kernels. The published AB-UPT accuracy uses surface and volume data, whereas the timed configuration uses surface data alone and omits domain-branched cross-attention.

12. PLAID Rotor37 — *live leaderboard, third place*

Rotor37 [41] is part of the PLAID benchmark suite (Safran, CC-BY-SA): 1,000 training and 200 test designs of a transonic axial compressor blade, 29,773 surface nodes each, with two operating scalars (rotational speed Ω , inlet pressure P) as input and three surface fields (density, pressure, temperature) plus three global quantities (mass flow, compression ratio, isentropic efficiency) as output. All six are scored by a server-side leaderboard.

In the 4 September 2026 snapshot, DIM-GP ranks **3rd of 16** at a total error of 0.000488 (displayed 0.0005), behind NPco (~0.00043) and PXTransolver (~0.00047) and ahead of Super-MARIO (~0.00077). The gap to NPco is approximately **0.00006 in aggregate RRMSE**, or **0.006 percentage points** when expressed as percentages (approximately 0.049% versus 0.043%).

RRMSE (lower is better)	Density	Pressure	Temp.	Massflow	Compr. ratio	Efficiency	total
NPco	0.0007	0.0007	0.0003	0.0003	0.0003	0.0003	0.0004
PXTransolver	0.0008	0.0008	0.0003	0.0003	0.0003	0.0003	0.0005
DIM-GP	0.0008	0.0008	0.0003	0.0003	0.0003	0.0003	0.0005
Super-MARIO	0.0013	0.0013	0.0005	0.0005	0.0005	0.0005	0.0008
MMGP	0.0031	0.003	0.0008	0.0005	0.0005	0.0005	0.0014
MeshFiLM	0.0027	0.0027	0.0007	0.0008	0.0009	0.0008	0.0014
MARIO	0.0035	0.0034	0.001	0.0008	0.0007	0.0006	0.0017
Baburu	0.0042	0.0042	0.0014	0.0007	0.0007	0.0009	0.002
Stealth	0.0029	0.0029	0.0009	0.0027	0.0027	0.0019	0.0023
ICLGS	0.0047	0.0047	0.0011	0.0021	0.0019	0.0014	0.0026
Vi-Transformer	0.0063	0.0062	0.0019	0.001	0.0011	0.0007	0.0029
Augur	0.0055	0.0053	0.0012	0.0028	0.0028	0.0019	0.0033
tp140205	0.0139	0.0136	0.0025	0.0047	0.0043	0.0042	0.0072
MGN	0.0114	0.0114	0.0024	0.0061	0.006	0.0071	0.0074
GeoFunFlow-3D	0.0319	0.0315	0.0134	0.0376	0.0383	0.0138	0.0278
FNO	0.084	0.0836	0.0086	0.0046	0.0042	0.0031	0.0313

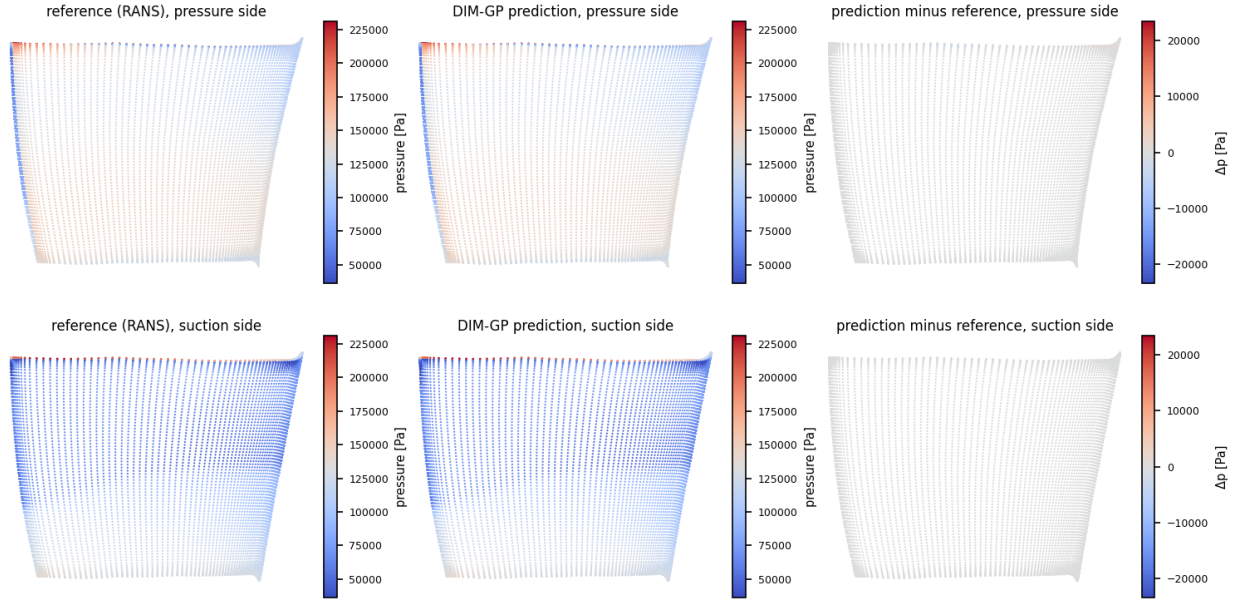


Figure 17. Rotor37, a held-out blade: surface pressure on both blade sides from the reference RANS solution, the DIM-GP prediction, and their difference.

13. PLAID VKI-LS59 — *live leaderboard, first place*

VKI-LS59 [41] is the suite’s 2D transonic turbine cascade: 671 training and 168 test profiles of a linear cascade (Safran, CC-BY-SA), 36,421 nodes each, solved by compressible RANS (Spalart–Allmaras) with the passage reaching Mach 1.78. Inputs are two operating scalars (inlet angle, outlet Mach number) and the mesh; the competition scores two nodal fields (Mach number and turbulent viscosity ν_t) and six global quantities (Q , power, the pressure and temperature ratios P_r and T_r , the isentropic efficiency η_{is} and the outlet angle).

RRMSE (lower is better)	nut	mach	Q	power	Pr	Tr	eth_is	angle_out	total
DIM-GP	0.0216	0.0093	0.0025	0.0062	0.0015	0	0.0306	0.0022	0.0093
PXTransolver	0.0205	0.0084	0.0039	0.0044	0.0015	0	0.0402	0.0023	0.0101
NPco	0.0243	0.0094	0.0025	0.0067	0.0018	0	0.0433	0.0019	0.0112
MARIO	0.0259	0.0112	0.0052	0.0077	0.0018	0	0.0453	0.0023	0.0124
ICLGS	0.0344	0.0155	0.0035	0.0092	0.0013	0	0.0375	0.0037	0.0131
MeshFiLM	0.0283	0.0102	0.0077	0.0091	0.002	0	0.0486	0.0025	0.0135
CRT	0.0337	0.0185	0.0015	0.0053	0.0018	0	0.0466	0.0023	0.0137
SAIR	0.0278	0.0122	0.0015	0.0049	0.0025	0	0.0586	0.0026	0.0138
Stealth	0.035	0.0176	0.0029	0.0067	0.0027	0	0.0699	0.0032	0.0173
gantnera	0.0329	0.0131	0.0109	0.0083	0.0026	0	0.0669	0.0037	0.0173
Vi-Transformer	0.0498	0.0232	0.0052	0.0083	0.0024	0	0.0621	0.0031	0.0193
MARIO (Samy)	0.0403	0.0208	0.0054	0.008	0.0029	0	0.0898	0.0033	0.0213
FNO	0.0846	0.018	0.0047	0.0062	0.0019	0	0.0539	0.0027	0.0215
test	0.0412	0.0186	0.0034	0.0052	0.0029	0	0.0976	0.0041	0.0216
Augur	0.0424	0.0221	0.012	0.0113	0.0027	0	0.0863	0.0045	0.0227
Rrrra	0.0423	0.0182	0.0189	0.0158	0.0038	0	0.0852	0.0071	0.0239
MMGP+	0.0822	0.0309	0.0023	0.0057	0.0026	0	0.1224	0.0033	0.0312
MGN	0.0771	0.0156	0.0716	0.0403	0.0064	0.0001	0.1625	0.0241	0.0497
transolver	0.0532	0.0139	1	1	1	1	1	1	0.7584
Naive_approach_GP	0.0669	0.0384	1	1	1	1	1	1	0.7632
akabalan_1fieldTesting	0.2849	0.0175	1	1	1	1	1	1	0.7878

The entry holds first place of 21 at 0.0093, approximately 8% lower aggregate error than PXTransolver. It leads the efficiency column and maintains competitive errors across the other scored outputs.

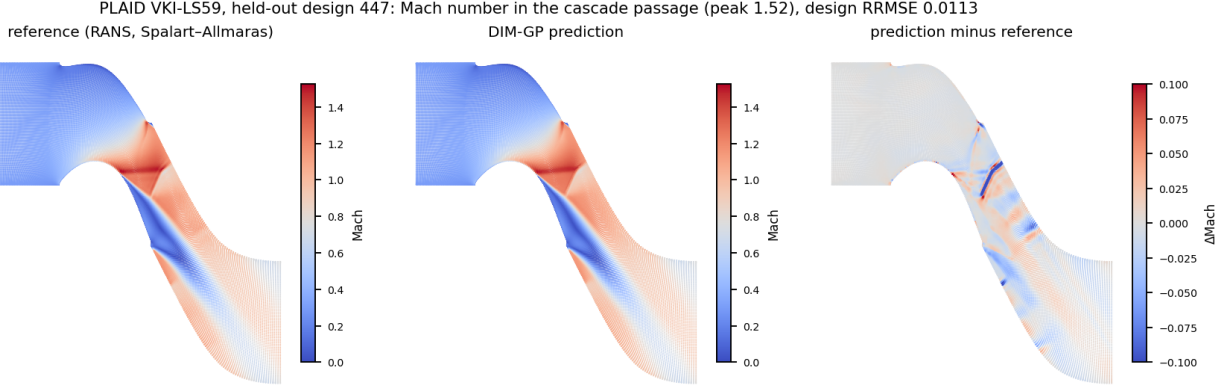


Figure 18. VKI-LS59, held-out cascade design 447: Mach number in the passage from the reference RANS solution, the DIM-GP prediction, and their difference (design RRMSE 0.0113). The passage shock and the suction-side expansion are placed within a few cells.

14. PLAID 2D Multiscale Hyperelasticity — *live leaderboard, first place*

This benchmark [41] is the suite’s variable-topology case: 764 training and 376 test representative volume elements of a porous hyperelastic material, each a different microstructure with five to eight holes (constant total porosity, node counts from 4,100 to 7,100), solved by FEniCS at finite strain. Inputs are three macroscopic strain components and the mesh; the competition scores seven nodal fields — the displacements u_1, u_2 , the four components of the first Piola–Kirchhoff stress, and the strain-energy density ψ — together with the homogenised effective energy. Every design has its own mesh topology.

RRMSE (lower is better)	u1	u2	P11	P12	P22	P21	psi	eff. energy	total
DIM-GP	0.003	0.0031	0.0093	0.0139	0.0095	0.0136	0.0263	0.002	0.0101
PXTransolver	0.0026	0.0028	0.0093	0.014	0.0095	0.0139	0.0262	0.0074	0.0107
test	0.0065	0.0064	0.0166	0.0256	0.017	0.0254	0.0267	0.0042	0.0161
NPco	0.0029	0.0028	0.0198	0.0285	0.0202	0.0283	0.0259	0.0037	0.0165
plaidtest	0.0068	0.007	0.0181	0.0278	0.0184	0.0275	0.0292	0.0057	0.0176
T&D	0.0059	0.0061	0.0201	0.0298	0.0205	0.0297	0.0266	0.0067	0.0182
Stealth	0.0055	0.0058	0.0231	0.033	0.0236	0.0329	0.0264	0.0039	0.0193
MiSe-GNN	0.012	0.014	0.0173	0.0312	0.0176	0.0314	0.0328	0.0111	0.0209
aravabt	0.006	0.0066	0.026	0.0368	0.0263	0.0364	0.0265	0.0055	0.0213
Augur	0.0109	0.0114	0.0208	0.0336	0.0212	0.033	0.0274	0.0188	0.0221
MeshFiLM	0.0091	0.0097	0.0261	0.0377	0.0267	0.0377	0.0281	0.008	0.0229
test	0.0119	0.0167	0.0262	0.0403	0.0267	0.0399	0.0295	0.0231	0.0268
FNO	0.0115	0.0117	0.0353	0.0513	0.0359	0.051	0.0329	0.012	0.0302
test1	0.0132	0.014	0.0329	0.0532	0.0326	0.0519	0.033	0.0117	0.0303
Vi-Transformer	0.0173	0.0172	0.0337	0.0581	0.0343	0.0571	0.0312	0.0113	0.0325
Unet	0.0291	0.0283	0.0349	0.0498	0.0347	0.05	0.035	0.018	0.035
Pmh	0.0161	0.0176	0.0366	0.0643	0.0367	0.0625	0.0362	0.0226	0.0366
ICLGS	0.0244	0.029	0.0473	0.0872	0.0482	0.0864	0.0433	0.0183	0.048
MARIO	0.0336	0.0377	0.0536	0.1067	0.0539	0.1053	0.0456	0.022	0.0573
GP	0.0748	0.075	0.0809	0.167	0.0815	0.1667	0.054	0.0216	0.0902
MuFi-MiSe	0.0108	0.0132	0.0305	0.0466	0.0307	0.0465	0.0315	1	0.1512
transolver	0.0129	0.0121	0.0327	0.0613	0.0332	0.0544	0.0335	1	0.155

The entry ranks first of 22 at 0.010093 (displayed 0.0101), approximately 6% below PXTransolver in aggregate error. The four stress errors are comparable to PXTransolver’s. DIM-GP has the

lowest effective-energy error, 0.0020, followed by NPco at 0.0037 and PXTransolver at 0.0074.

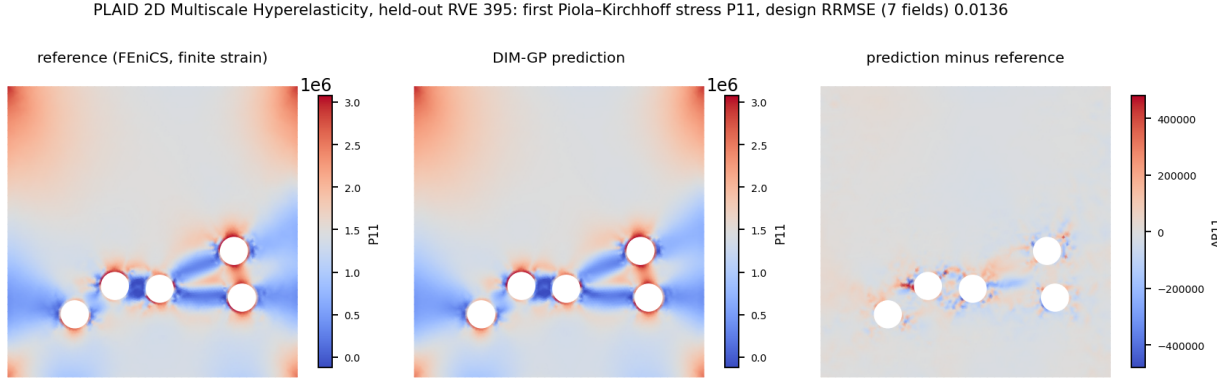


Figure 19. 2D Multiscale Hyperelasticity, held-out RVE 395 (five holes): first Piola–Kirchhoff stress P_{11} from the FEniCS reference, the DIM-GP prediction, and their difference (design RRMSE over the seven fields 0.0136). The stress concentrations between the holes are reproduced in place.

15. PLAID Tensile2d — *live leaderboard, second place behind a declared solver*

Tensile2d [41] is the suite’s entry-level structural case: a 2D plane-strain, quasi-static, small-strain elastoplastic specimen under a uniform top traction, meshed once per design with 6,000–10,000 nodes (Safran, Z-set solver). Inputs are six scalars — the load P , four parameters $p_1 \dots p_4$ of the isotropic hardening law and the Young’s modulus p_5 — and the mesh; the competition scores five nodal fields (U_1 , U_2 , σ_{11} , σ_{22} , σ_{12}) and three scalars (the maximum von Mises stress, and the maximum U_2 and σ_{22} on the top edge). 500 designs carry labels, 200 form the test set.

RRMSE (lower is better)	U1	U2	sig11	sig22	sig12	max vM	max U2 top	max sig22 top	total
PhysicsX Agent (reverse-engineered)	0	0	0	0	0	0	0	0	0
DIM-GP	0.00040	0.00050	0.0013	0.0006	0.0012	0.0004	0.001	0.0004	0.0007
NPco	0.0002	0.0003	0.0013	0.0006	0.0009	0.0044	0.0006	0.0016	0.0013
PhysicsX Transolver	0.0002	0.0002	0.0012	0.0006	0.0009	0.0046	0.0008	0.0017	0.0013
Stealth	0.0003	0.0003	0.0012	0.0006	0.0009	0.0048	0.0008	0.0016	0.0013
MeshFiLM	0.0002	0.0003	0.0012	0.0006	0.0009	0.005	0.0022	0.0015	0.0015
plaidtest	0.0013	0.0017	0.0029	0.0013	0.0018	0.0051	0.003	0.0017	0.0023
PassionTraining	0.0009	0.0011	0.0021	0.001	0.0016	0.0066	0.0044	0.0019	0.0025
MMGP	0.0015	0.0009	0.0031	0.0013	0.0021	0.005	0.0053	0.0017	0.0026
test	0.0015	0.0022	0.0036	0.0016	0.0024	0.0069	0.0025	0.0019	0.0028
ICLGS	0.0028	0.004	0.0048	0.0018	0.0032	0.0065	0.0021	0.0018	0.0034
CRT	0.0013	0.0015	0.0057	0.0022	0.0039	0.0079	0.0024	0.0026	0.0034
sangmin12312343	0.0018	0.0018	0.003	0.0019	0.0023	0.0108	0.0066	0.0016	0.0037
MARIO	0.0023	0.003	0.004	0.0017	0.0023	0.0088	0.0063	0.0023	0.0038
MMVT	0.0043	0.0051	0.0089	0.0037	0.0051	0.0068	0.0073	0.0019	0.0054
Augur	0.0037	0.0048	0.0081	0.0035	0.005	0.0101	0.014	0.0034	0.0066
MGN	0.0034	0.0043	0.0047	0.0013	0.0016	0.0169	0.0292	0.0022	0.008
Vi-Transformer	0.0086	0.0091	0.0184	0.0102	0.0146	0.009	0.0203	0.0021	0.0116
FNO	0.0174	0.011	0.025	0.0057	0.0135	0.0085	0.0152	0.0021	0.0123
JBone	0.3054	0.3891	0.4022	0.2354	0.268	0.0052	0.0017	0.0017	0.2011
MuFi-MiSe	0.0023	0.0036	0.0042	0.002	0.0024	1	1	1	0.3768
transolver	0.0029	0.0036	0.0049	0.0016	0.0022	1	1	1	0.3769

DIM-GP ranks second of 22 at a displayed aggregate error of 0.0007. In its article *How an AI Agent*

Cracked an Industry Benchmark, PhysicsX reports that AI agents reconstructed the reference simulator by inferring the constitutive law and simulation setup from the supplied training data [44]. Its entry ranks first with zero error at the leaderboard’s displayed precision.

The DIM-GP workflow runs on the CPU in minutes.

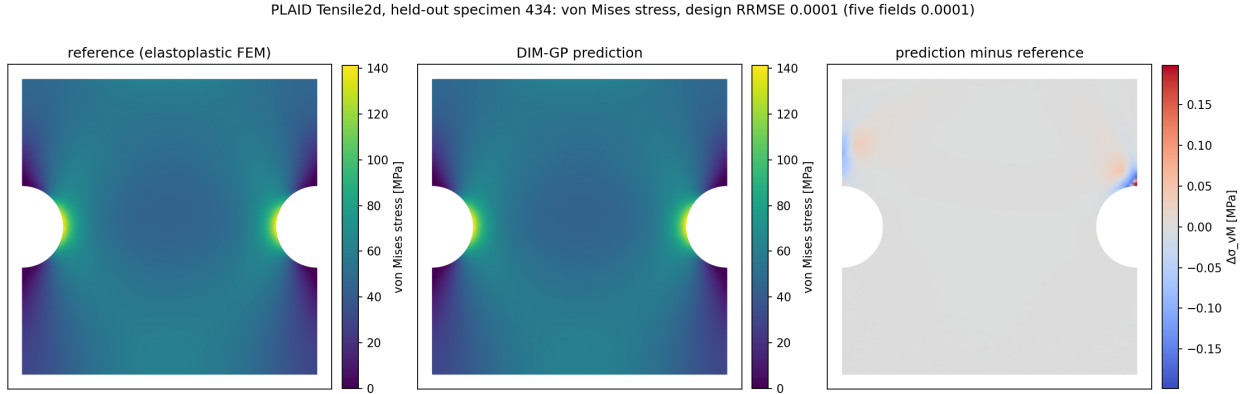


Figure 20. Tensile2d, a held-out specimen: von Mises stress from the reference elastoplastic solution, the DIM-GP prediction, and their difference.

16. PLAID 2D ElastoPlastoDynamics — *live leaderboard, first place*

This is a transient solid-mechanics case [41]: a 200×100 mm steel plate with up to four holes and edge notches, clamped on the left and pulled at 500 mm/s on the right, computed by the benchmark’s authors with an explicit finite-element solver, a non-linear, non-local constitutive law and element erosion, and delivered as 41 time steps of the two displacement fields on 19,000–30,000 nodes per design. There are no input scalars: the geometry is the only variable, 1,000 designs are labelled and 18 form the test set. The competition scores the full $U_x(t)$ and $U_y(t)$ trajectories.

Transient field prediction substantially increases the scale of the learning problem. Each design contains a sequence of complete mesh fields rather than a single steady-state solution. Here, 41 time steps across 19,000–30,000 nodes produce approximately 0.8–1.2 million node-time states per design and output field. Memory use and training cost therefore grow rapidly with both spatial resolution and sequence length, particularly for architectures that represent mesh states as large token sequences. This makes the benchmark a demanding test of an architecture’s ability to model high-resolution spatial fields and their evolution over time.

RRMSE (lower is better)	U_x	U_y	total
DIM-GP	0.0005	0.0167	0.0086
test	0.0008	0.017	0.0089
DAFNO	0.0025	0.0291	0.0158
FNO	0.0031	0.0399	0.0215
Vi-Transformer	0.0186	0.0269	0.0227
MGN	0.0073	0.0403	0.0238
MARIO	0.0059	0.058	0.0319
Augur	0.0264	0.0427	0.0346

The entry holds first place of eight at 0.008605, with the lowest displayed U_x and U_y errors in the archived table. The displayed U_x values are 0.0005 for DIM-GP and 0.0008 for the next entry.

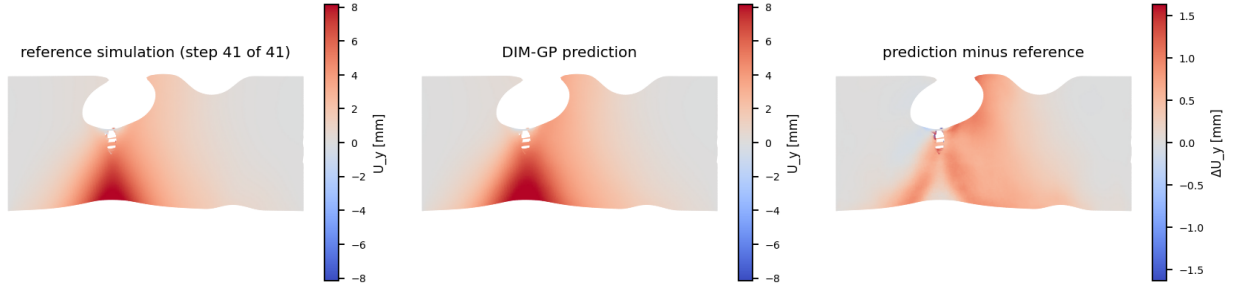


Figure 21. 2D ElastoPlastoDynamics, held-out design 8 drawn in its deformed shape at the final step: vertical displacement U_y from the reference simulation, the DIM-GP prediction, and their difference (design RRMSE U_x 0.0099, U_y 0.0197).

The board metric normalises each field by its maximum over the whole test set, with constants of roughly 340 mm (U_x) and 25 mm (U_y).

17. PLAID 2D_profile — *live leaderboard, third place*

2D_profile [41] is the suite’s shape-only aerodynamics case: 300 training and 100 test airfoil profiles in a transonic flow, each with its own triangular mesh of 35,000–39,000 nodes cut close to the profile, and **no input scalars at all** — the geometry is the whole input, with the effective angle of attack baked into the shape. The competition scores four nodal fields (Mach number, pressure and the two velocity components). The test cases include transonic shocks.

RRMSE (lower is better)	Mach	Pressure	Velocity-x	Velocity-y	total
PXTransolver	0.0134	0.0108	0.0155	0.0139	0.0134
NPco	0.0148	0.0097	0.0172	0.0136	0.0138
DIM-GP	0.0167	0.011	0.0196	0.0134	0.0152
test-ts-dino	0.0161	0.0114	0.0184	0.0149	0.0152
Stest	0.0168	0.0111	0.0198	0.0135	0.0153
plaidtest	0.0163	0.0111	0.0197	0.0146	0.0154
Stealth	0.0166	0.011	0.0195	0.0154	0.0157
MM-GP+	0.0208	0.0139	0.0243	0.016	0.0187
Transolver+	0.0209	0.0131	0.0267	0.019	0.0199
test	0.0211	0.0154	0.0242	0.0192	0.02
Super MARIO	0.022	0.0158	0.0251	0.0196	0.0206
MARIO	0.023	0.0146	0.0264	0.021	0.0213
MeshFiLM	0.0265	0.0177	0.0316	0.0243	0.025
CRT	0.0278	0.0171	0.0323	0.0244	0.0254
MuFi-MiSe	0.0279	0.018	0.0328	0.0315	0.0276
MiSe-GNN	0.0309	0.0212	0.0358	0.026	0.0285
Vi-Transformer	0.036	0.0167	0.0403	0.0307	0.0309
MMGP	0.0439	0.0208	0.0471	0.0342	0.0365
Augur	0.0469	0.0248	0.0538	0.0445	0.0425
ICLGS	0.0777	0.0248	0.0885	0.0601	0.0628
FNO	0.0988	0.0785	0.1148	0.0967	0.0972
elzigomario	0.1893	0.1004	0.2295	0.1199	0.1598

DIM-GP ranks third of 22 in the 7 September 2026 snapshot, with an aggregate error of **0.0152** versus **PXTransolver’s 0.0134**. The numerical gap is **0.0018**, equivalent to **0.18 percentage points** when these relative errors are expressed as percentages (1.52% versus 1.34%). DIM-GP’s Velocity-y error of 0.0134 is the lowest displayed value on the board; its pressure error of 0.011 is joint-third.

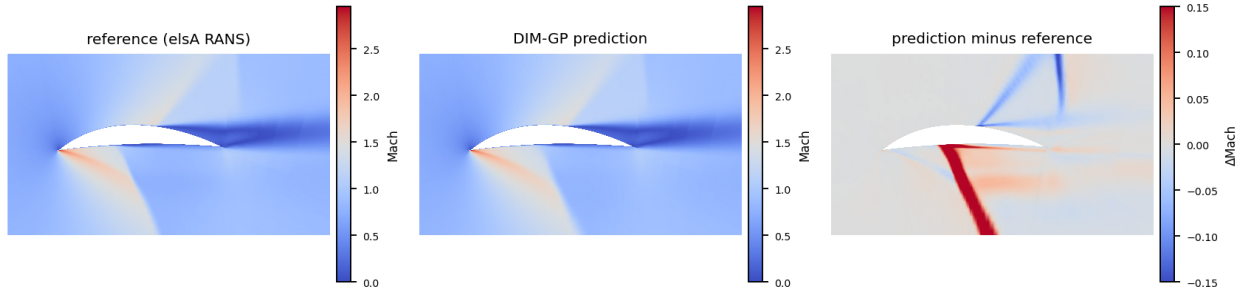


Figure 22. 2D_profile, held-out design 272: Mach number from the elsA reference, the DIM-GP prediction, and their difference (design RRMSE 0.0192, peak Mach 2.95). The prediction reproduces the leading-edge supersonic pocket and shocks; the largest differences occur near the shock fronts.

18. Bayesian optimization — *concrete mix design, six approaches, one virtual problem*

Bayesian optimization uses predictions and uncertainty estimates to select the next design to evaluate. Here, STOCHOS searches for concrete recipes that combine high 28-day compressive strength with low cradle-to-gate CO₂ and material cost. Seven ingredient masses per cubic metre are varied, subject to a total mass of 2,195–2,551 kg.

The study uses a fixed virtual test bench fitted to the 1,022 usable rows of the public Concrete Compressive Strength dataset (UCI, 1,030 laboratory recipes) [45]. It predicts strength; CO₂ and cost follow from the ingredient quantities, sourced emission factors and an indicative price basket. Six approaches receive the same starting recipes and a budget of 60 evaluations: STOCHOS, BoTorch’s qNEHVI [46], BayBE’s desirability approach [47], a random-forest optimizer, D-optimal design of experiments and random search.

HOW BAYESIAN OPTIMIZATION WORKS

Each evaluation improves the information used to select the next candidate.

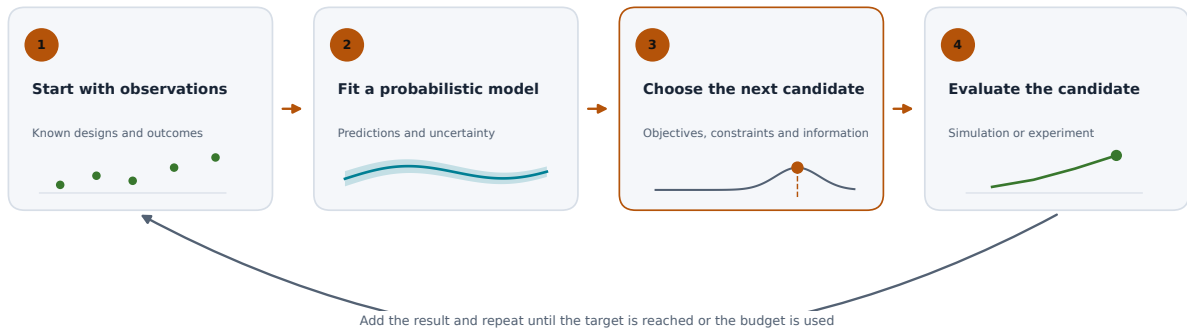


Figure 23. The Bayesian-optimization loop. Existing observations train a probabilistic model, which estimates outcomes and uncertainty. An acquisition decision selects a promising candidate, the candidate is evaluated, and the result updates the next iteration. In this study, the candidates are concrete recipes and the evaluation is performed by the fixed virtual test bench.

Part A — reaching a specification. Five levels of increasing difficulty each demand a minimum strength together with a CO₂ and a cost ceiling on the *same* recipe; difficulty is characterized by an estimated probability that a random valid recipe qualifies (1 in 9 for S1 down to 1 in 66,667 for S4.5), and an unreachable sixth level serves as a control that no method passes. Ten repeated runs per level, at round sizes of one, three

and six recipes. The table counts, per level, how many of the 10 runs at one recipe per round found a qualifying recipe.

Runs that reached the specification (of 10)	S1	S2	S3	S4	S4.5
STOCHOS	10	10	10	10	10
BoTorch (qNEHVI)	10	10	10	10	8
BayBE	10	10	10	9	2
Random forest	10	10	7	3	1
Classical DoE	10	10	6	0	0
Random search	10	7	4	1	0

Over all five levels and all three round sizes **STOCHOS reaches the specification in 149 of 150 runs**, against 137 for BoTorch, 111 for BayBE, 87 for the random forest, 78 for classical DoE and 66 for random search; it is the only method that solves the extreme level in every sequential run. Among runs in which both STOCHOS and BoTorch succeeded, STOCHOS reached the target sooner in 99 of 136 comparisons, later in 16 and at the same experiment in 21 (reported two-sided sign test $p = 8.6 \times 10^{-16}$).

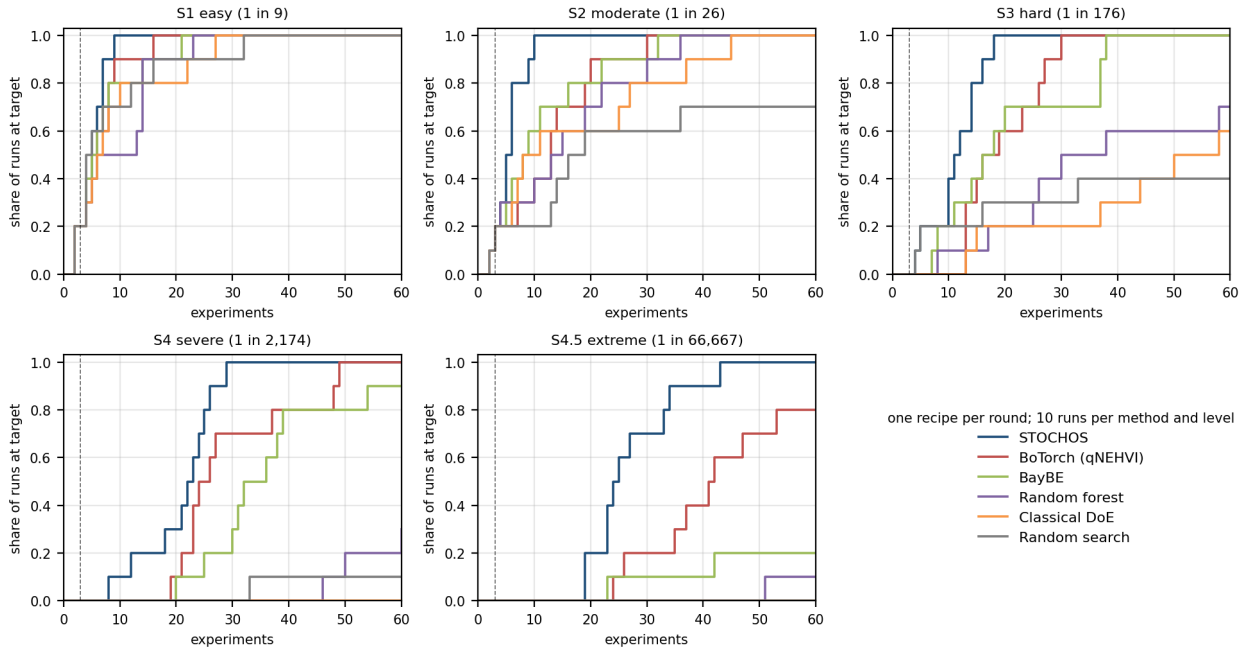


Figure 24. How quickly each method reaches the specification: the share of the 10 repeated runs that had found a qualifying recipe by a given experiment number, one recipe per round, for the five difficulty levels of Part A. The first three recipes are shared starting points (dashed line); a curve that rises earlier reached the target sooner.

Part B — discovering the trade-off set. With no single target, a method is scored on how much of the achievable trade-off region it discovers: the hypervolume of its evaluated recipes against a frozen reference point, averaged over 30 repeated runs; the table gives the coverage after 60 experiments at round sizes of one, five and ten.

Trade-off coverage after 60 experiments (hypervolume fraction)	1 / round	5 / round	10 / round
STOCHOS	0.9225	0.9150	0.8822
BoTorch (qNEHVI)	0.9105	0.9045	0.8917
BayBE	0.8900	0.8838	0.8726
Random forest	0.7740	0.7785	0.7679
Classical DoE	0.7345	0.7345	0.7345
Random search	0.6840	0.6840	0.6840

STOCHOS leads at round sizes one and five and is second at ten, where the batch of ten commits a sixth of the budget before any result returns. Choosing one recipe at a time, it reaches 80% coverage in a median of 32 experiments and 90% in 46 (on 20 of 30 runs, the most of any method); random forest, classical DoE and random search never reach 80% inside the budget. The extended-budget study of the classical design makes the gap concrete: a D-optimal plan needs 180 experiments to reach the 70% coverage that STOCHOS passes at 21.

Cost of the decisions. Part B’s sequential runs have median elapsed times of **26.7 minutes for STOCHOS**, **50.2 minutes for BoTorch** and **147.6 minutes for BayBE** over 30 runs per method, each with a 60-experiment budget. STOCHOS uses approximately **47% less elapsed time than BoTorch** in this comparison, measured for model fitting and proposal generation in the virtual benchmark. The methods with the lowest computational overhead, classical DoE and random search, have lower reported coverage.

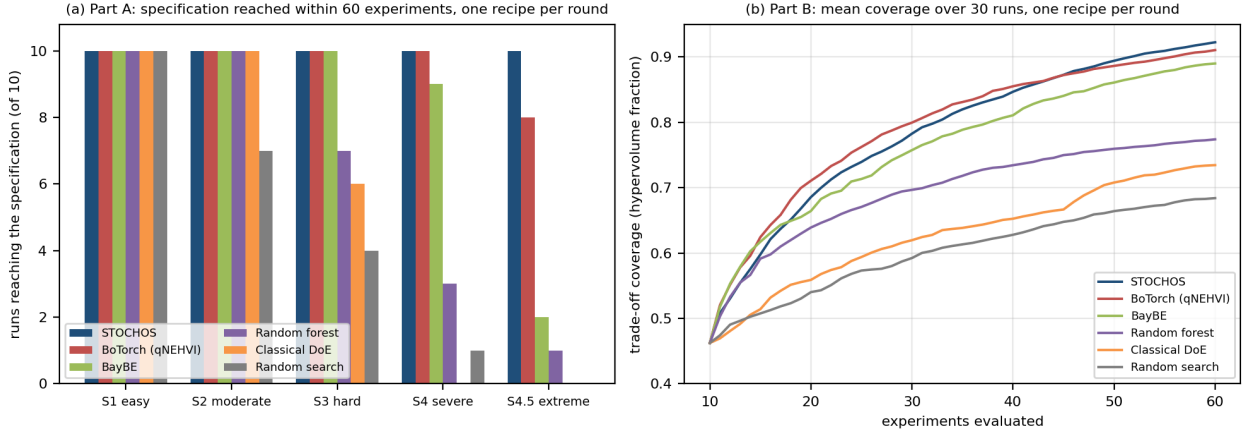


Figure 25. Bayesian optimization on concrete mix design, six methods on one problem with a budget of 60 experiments. (a) Part A, one recipe per round: number of the 10 repeated runs that found a recipe meeting the specification, per difficulty level. (b) Part B, one recipe per round: mean trade-off coverage (hypervolume fraction against the frozen reference point) after each evaluated experiment, 30 runs per method.

Appendix A. Sources and authorship

Public leaderboard rows, published baseline values and internal experiments are distinguished in the relevant sections. Numerical ranks refer to the stated comparison fields and snapshot dates. PLAID scores are computed by the benchmark platform; STOCHOS training and timing measurements are reported by the authors.

This report is authored by PI Probaligence GmbH, the developer of STOCHOS. References link to the benchmark datasets, published methods and evaluation protocols used in the comparisons.

References

References identify benchmark datasets, evaluation procedures, methods and author descriptions. For tabular entries, registered TabArena references were used where available.

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